JÖNKÖPING UNIVERSITY
School of Engineering

Shaping Airport Mobility:

Human-Centred Design of an Autonomous Wheelchair for People with Reduced Mobility

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This thesis work has been carried out at DAAV S.A. in Switzerland. It is part of the two-year university diploma program within the subject area of Product Design in the School of Engineering of Jönköping University.

The author takes full responsibility for opinions, conclusions and findings presented.

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Abstract

Airports are experiencing a growing demand for mobility assistance, with Spanish airports alone handling more than 6,600 assistance requests per day in 2024, all the while relying on manual labour and conventional wheelchairs. Although autonomous wheelchairs are emerging in the airport market, current solutions focus primarily on technical performance rather than the user's needs. Therefore, this thesis aimed to identify the key requirements for the development of an autonomous wheelchair for people with reduced mobility in airport environments and to explore how these requirements can be translated into the design of such a wheelchair.

The study followed the Double Diamond framework and adopted a human-centred design approach to understand the experiences, needs, and challenges of people with reduced mobility in airport environments. The methods and steps taken included stakeholder analysis, competitor analysis, observations, semi-structured interviews, contextual airport analysis, user journey mapping, thematic analysis, brainstorming, CAD modelling, and prototyping. A user validation survey was conducted to evaluate the final design proposal and collect feedback regarding usability, accessibility, comfort, and visual appearance.

The study demonstrated that autonomous wheelchairs should be designed not only as technology-focused navigation systems but also as user-centred mobility products that support independence, comfort, accessibility, and dignity. The findings identified key requirements for autonomous wheelchairs in airport environments and were translated into a final design proposal that integrated luggage and mobility-aid storage, accessible transfer solutions, ergonomic features, and a modern visual language adapted to airports. These findings contribute to the limited body of user-centred and design-oriented research on autonomous wheelchairs for airports.

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1 Introduction

This chapter introduces the background and context of the study, outlines the problem addressed, and presents the purpose and research questions. It further defines the delimitations of the study to clarify its scope and focus.

1.1 Background

Air travel has skyrocketed over the past decades. In 2024 alone, approximately 1.1 billion people were travelling by air in the European Union (Eurostat, 2025). At the same time, the World Health Organization (2019) estimates that around 15% of the world's population lives with some form of disability, whether visible or invisible. This number is expected to rise as populations age, with disability rates being higher among the elderly, as a result of health risks, injuries or chronic illnesses (United Nations, n.d.).

Within the aviation context, passengers who require additional assistance are commonly referred to as passengers with reduced mobility (PRM). This group includes individuals who may have temporary or permanent mobility limitations, such as elderly passengers, people with reduced walking endurance, and individuals with physical, sensory, or cognitive impairments. PRM passengers represent a substantial and growing share of airport activity. For example, Spanish airports collectively handled more than 6 600 PRM assistance requests per day in 2024 (Aena, 2024). Similarly, across the UK aviation network, approximately 1.9% of total passengers, which is equivalent to 5.5 million people, requested PRM assistance in 2024 (Civil Aviation Authority, 2025).

PRM assistance services heavily rely on complex coordination of human labour and mobility aids. Yan et al. (2025) mention that the scheduling and allocation of staff impacts service performance. Poor workforce planning has been seen to lead to excessive wait times, reduced service quality, and dissatisfaction among PRM passengers (Grahn & Jacquillat, 2020). Moreover, PRM service staff have physically demanding jobs, difficult working hours and often face relatively low wages, contributing to a high employee turnover. Inefficient management of PRM services can even cause delays in downstream operations such as boarding. These challenges highlight how the influence of accessibility services extends beyond individual passenger experience to overall airport efficiency.

In response to these challenges, autonomous mobility technologies are increasingly explored within airport environments. Autonomous wheelchairs in particular, have the potential to reduce reliance on manual assistance, improve service efficiency, and increase independence for PRM passengers. By allowing the passengers to navigate autonomously within the terminals, these products can reduce staff workload, lower long-term operational costs for airports, and offer PRM passengers faster access to mobility services.

At the same time, the introduction of autonomous wheelchairs into public airport environments extends beyond technical functionality alone. Airports are crowded, unfamiliar, and often stressful places in which users must quickly understand and trust the systems surrounding them. In this context, the success of autonomous wheelchairs depends not only on their software and navigation performance, but also on how the

wheelchairs are experienced by users physically, cognitively, and emotionally. Factors such as comfort, accessibility, aesthetics, and overall usability are therefore important considerations in the development process. As autonomous wheelchairs continue to evolve, there is an increasing need to understand how such requirements can inform their design in a way that is relevant for airport environments and meaningful for the people expected to use them.

1.2 Project context

This thesis was conducted as part of a five-month internship for DAAV S.A. (DAAV), a Swiss company developing autonomous wheelchairs for airport environments (DAAV, 2026). This internship provided the opportunity to work closely with a multidisciplinary team involved in the development of autonomous mobility solutions for passengers with reduced mobility.

Working from the company's office in Switzerland enabled continuous discussions around the project and provided valuable learning opportunities through guidance from all the team members. These interactions contributed to technical insights throughout the development of the thesis project.

Alongside the research and design proposals in this thesis, the internship also involved the development of a new version of the DAAV B.AIR wheelchair with another thesis student from KTH University. Working on this project provided a deeper understanding of the wheelchair's mechanical constraints, component integration, and manufacturing considerations, as this model was produced and implemented. This experience informed the design decisions made within this thesis and helped ensure that the proposed concepts remained grounded in realistic technical and functional requirements.

1.3 Problem statement

While the interest in autonomous wheelchairs for airport environments is increasing, most current options seem to prioritise technical performance, navigation accuracy, and operational efficiency. Although these aspects are essential for autonomous functionality, they alone do not determine whether a product can successfully support the needs of passengers with reduced mobility in real airport contexts.

For PRM passengers, mobility challenges are not only limited to transportation. They also involve questions of independence, comfort or accessibility when interacting with mobility aids in public settings. An autonomous wheelchair that is difficult to understand, visually intimidating, overly medical, or poorly adapted to airports may negatively influence the user experience regardless of its technical performance. Similarly, human-centred design factors such as ergonomics, inclusive features, and visual appearance may all affect how appropriate and usable the wheelchair is within airport environments.

Existing research on autonomous mobility systems often focuses on engineering, robotics, or navigation performance, while user-centred and design-oriented perspectives are limited, particularly during the early stages of development and

requirements definition (Chavarria et al., 2026). There is a lack of research dedicated to identifying which requirements should guide the design development of autonomous wheelchairs specifically for airport use.

This gap suggests a need for a human-centred study that identifies and translates relevant user needs, as well as environmental and functional requirements, into the exterior design of autonomous wheelchairs for airport contexts. Addressing this gap may contribute to a broader understanding of how autonomous wheelchairs can support usability, accessibility, independence, and positive user experiences for people with reduced mobility travelling through airports.

1.4 Purpose and research questions

Autonomous wheelchairs have the potential to positively reshape airport PRM services by supporting more independent and flexible mobility for passengers with reduced mobility. As described above (1.3), in order to function effectively in real airport environments, the wheelchair's development must consider more than just software and technical performance. It must also take into account the physical, environmental, and experiential conditions that influence how the product is used and perceived within this context.

Therefore, the purpose of this thesis is to identify key requirements for the development of an autonomous wheelchair for people with reduced mobility in airport environments, and to explore how these requirements can be translated into the design of such a product. Later these findings are implemented to design an autonomous wheelchair for airports.

This thesis follows a human-centred design approach, involving users from the early design development steps, alongside functional requirements from airports and the company itself.

This project primarily considers the most common PRM groups in airports, mainly elderly passengers, individuals who cannot walk long distances and those who require assistance navigating complex airport environments, that retain enough cognitive and physical abilities to use an autonomous wheelchair independently.

Through research, analysis, and design, the study aims to contribute to the central research problem of how autonomous wheelchairs can be designed to better support PRM users in airport environments.

To analyse this research problem, the thesis is guided by the following two research questions:

RQ1: What are the key requirements for the development of an autonomous wheelchair for people with reduced mobility in airport environments?

RQ2: How can these requirements be translated into the design of an autonomous wheelchair for PRM users in airport environments?

1.5 Delimitations

A vast number of user needs are at stake when developing a product, especially for PRM users. To limit the scope of this thesis, the user needs chosen to develop further in this study were the most frequent ones mentioned and collected during the semi-structured interviews and observations. As for the design itself, this project focuses solely on the exterior hardware design of an autonomous wheelchair, without analysing any manufacturing and production cost considerations that may come into play.

1.6 Use of artificial intelligence

Artificial intelligence was used to improve the text structure and formulation in certain areas of the report, however all analysis and critical thinking in this research was conducted by the author.

2 Theoretical framework

This chapter provides a theoretical basis for the study.

2.1 Human-centred design

Human-centred design (HCD) is an approach that places users, their needs, and their experiences at the centre of the design process. Rather than designing purely around technical details, HCD aims to develop solutions and products that are truly meaningful and usable for the users. HCD emphasises empathy and user involvement during the development process. Hassan et al. (2023) mention that a user-centred approach leads to better understanding users' needs and behaviours, which can create more efficient products.

Similarly, Silberer et al. (2023) argue that understanding user needs is a success factor for new innovations and can significantly impact whether a product will be rejected or accepted.

2.2 Usability

Usability is part of human-centred design and, according to Buker et al. (2022), represents the utility of a product not only in a functional way, but also in a cognitive way. It means that a product should both function correctly and feel intuitive and efficient to use. Rubin (2008) also mentions that a truly usable product allows users to perform tasks naturally and without frustration, hesitation, or confusion.

2.3 User interaction

User interaction focuses on the way people communicate and engage with products, systems, and interfaces. In product design, interaction goes beyond physical functions, it also includes emotional responses, interpretation, and reactions during the use of a product. Chamorro-Koc et al. (2009) explain that human experience strongly shapes how users interact with a product, for instance cultural background and previous experiences can influence people's understanding of its function. Because of this, interaction design should not only support task completion but also create clarity and confidence for users.

Visual and aesthetic qualities also influence how users perceive interaction. A product's appearance can affect apparent usability, meaning that users often judge how easy to use a product seems to be before interacting with it directly (Mugge & Schoormans, 2012). Attractive and visually understandable products are therefore more likely to appear intuitive and approachable.

2.4 Ergonomics

According to the International Ergonomics Association (IEA) (IEA, 2000), ergonomics applies theory, principles, and methods to design in order to optimise both users' well-being and overall performance of the product. This discipline considers users' capabilities and limitations to positively influence comfort, safety, and efficiency. Ergonomics aims to ensure that products are adapted to physical characteristics of the users which can help minimise strain, risk of injury, and contribute to a more comfortable user experience.

2.4.1 Anthropometrics

Anthropometrics refers to the measurement of human body dimensions and their proportions when designing products. These measurements are often used to make sure that products accommodate a broad range of users safely and comfortably. In ergonomics, anthropometric data helps determine dimensions such as seat width, armrest height, footrest width, and many others. Pheasant and Haslegrave (2006) suggest that anthropometric design aims to adapt products' sizes to human variability rather than forcing users to adapt themselves to the product.

2.5 Inclusive and universal design

Inclusive design aims to create products and services that can be accessed by a wide range of people, regardless of age or ability. Inclusive design recognises that human needs and capabilities are varied and may change over time. According to the Design Council (n.d.), inclusive design should place people at the centre of the design process while acknowledging diversity and differences. It does not mean that a product should meet every need, but that it should limit exclusion by finding solutions that can benefit many users. The principles of inclusive design also extend beyond only physical accessibility. Inclusive products should be intuitive and convenient, so people can use them easily and with dignity (Commission for Architecture and the Built Environment (CABE), 2006).

A closely related concept, which often overlaps, is universal design. Universal design aims to minimise barriers and accommodate user needs though proactive design decisions. A significantly recognised framework within universal design is the Seven Principles of Universal Design: equitable use, flexibility in use, simple and intuitive use, perceptible information, tolerance for error, low physical effort, and size and space for approach and use (Centre for Excellence in Universal Design (CEUD), n.d.). Together, these principles provide guidance for developing products that are accessible, understandable, comfortable, and usable for a broad spectrum of users.

The principles of universal design are often used to support inclusive design by translating broad accessibility goals into practical design considerations. For example, simple and intuitive use encourages products to be easy to understand regardless of prior experience, while low physical effort and adequate space for approach accommodate users with varying physical capabilities (CEUD, n.d.). These principles can therefore serve as a useful framework for evaluating and informing design decisions throughout the product development process.

2.6 Autonomous wheelchairs

Autonomous wheelchairs are smart mobility systems designed to support users with reduced mobility by using autonomous navigation and assistive technologies. They integrate sensors, software systems, and obstacle detection to create a safe travel experience for the users, so that no constant intervention is needed from the user. These wheelchairs aim to increase independence, reduce physical and cognitive strain, and improve safety during transportation (Simpson, 2005).

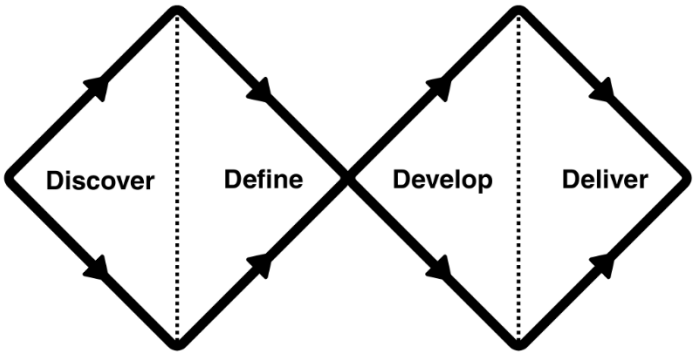
3 Methods

The chapter provides an overview of the study's research methods.

3.1 Double Diamond Framework

The Double Diamond is a design framework used around the world and developed by the British Design Council. It aims to guide a designer through various phases of a design process, from divergent to convergent thinking, which, as visualised in Figure 1, results in the shape of two diamonds (British Design Council, n.d.).

Figure 1: *The Double Diamond Framework*



This framework consists of four phases: Discover, Define, Develop, and Deliver (Majka, 2024).

- Discover: During the first phase, research is conducted to collect information about a specific problem in order to understand it, without directly assuming what that problem is. It can be done through research, observations, market analysis, collection of data and by interacting with the users.

- Define: The goal of the second phase is to analyse and synthesise the data collected in the previous step. It allows to narrow down the problem and to establish the project objectives, making sure it aligns with user needs.

- Develop: This phase involves creating many solutions to solve the problem. It can be done through brainstorming, ideation, and concept development, steps where ideas are shared and explored, as well as prototypes that are used to test those solutions.

- Deliver: The final phase includes more in-depth tests, where the final solution is chosen and developed further into a final concept.

3.2 Discover

3.2.1 Stakeholders

Identifying key stakeholders clarifies which groups will utilize the future product, contribute to its production, or manage the resulting service (Tideman, 2008). Each

stakeholder maintains its own objectives and criteria for success. Therefore, the designer has to determine their levels of influence and compromise on their preferences (Aikhuele, 2019). Evaluating project challenges across diverse stakeholder roles ensures a comprehensive understanding of the criteria for a successful design and generates valuable insights (Hall, 2013).

3.2.2 Competitor analysis

Competitor analysis is a vital part of the design research that informs decision-making and aligns product proposals with actual market conditions and user expectations. Designers determine who their main competitors are and which products they offer. They identify the strengths and weaknesses of these products, evaluate their features and user experience. This analysis indicates possible unmet needs and gaps in the market. It helps to avoid repeating past industry errors as well as to pinpoint areas where a new product can add unique value. Summarizing key findings and challenges provides the foundation for a competitive product design (Milton & Rodgers, 2013).

3.3 Define

3.3.1 Persona

Persona is a method where the designer creates an archetypal user that represents the targeted users for a product or service (Grzybowska, 2025). A persona can describe a user's background, needs, frustrations and interests that represent and resonate with a larger group of users.

The most significant benefit of personas is their ability to focus product design teams on the actual goals and needs of the target customers. According to Miaskiewicz and Kozar (2011), personas bring the target consumers to life and help to integrate their needs and goals as a central driver of design process.

3.3.2 Try it yourself

The "Try it yourself" method has become increasingly popular. The designer acts as the user by testing the product or prototype to better understand the challenges of the users and be able to identify issues rapidly. This approach promotes empathy and can allow to expand the design solutions (Milton & Rodgers, 2013).

Although this approach leads to identifying new perspectives or needs, it is important to also utilize other methods to collect user needs, as this method can bring biased results, especially if the designer does not have the same characteristics as the users. This bias concern is heightened when the targeted users have reduced mobility or disabilities.

3.3.3 Observations

Conducting observations with users is a key design methodology used to uncover unspoken needs and motivations. Some user needs are implicit; users take them for granted and therefore do not actively report them in surveys or interviews. Users are often unable to reliably remember or clearly describe their own behaviour. They may exaggerate or under-represent certain challenges. Observations help uncover these invisible points that the user has stopped noticing and provide a neutral account of the user experience. This technique allows the designer to observe and document everything users do, along with the context and environment of their interactions (Säfsten & Gustavsson, 2020).

3.3.4 Semi-structured interviews

Semi-structured interviews usually rely on an interview guide containing questions designed to meet the research objectives, but do not follow it rigidly or in the same sequence for every interview. It serves as a flexible framework that helps guide and shape the natural flow of each conversation. Instead of having specific options for answers as seen in surveys, semi-structured interviews often use open-ended questions to better capture context and qualitative details. A key advantage of semi-structured interviews is that they maintain a clear focus while allowing the researcher the flexibility to explore relevant ideas that emerge during the discussion, leading to a deeper understanding of the product or service being evaluated (Adeoye-Olatunde & Olenik, 2021)

3.3.5 User journey

User journey mapping is a methodology that visually represents the sequence of interactions and experiences users encounter when engaging with a product or service. This method helps designers better understand user behaviours and identify pain points or possible improvements for the user experience. Writing down steps of the interaction process, the user journey helps understand how users navigate and perceive products or systems when used in their environment. It is particularly useful in user-centred design projects as it combines functional actions with emotional responses, which can allow designers to discover frustrations or unmet needs that could arise (Ly et al., 2021).

3.4 Develop

3.4.1 Brainstorming

Brainstorming serves to generate ideas, find solutions to specific problems and support conceptual design. It aims to generate high volumes of ideas to increase the likelihood that some might bring promising results. By leaving aside limitations and judgement, more creative ideas can be produced in a large quantity before being narrowed down and selected in future phases (Wilson, 2013).

3.4.2 Keywords and mood board

Selecting keywords can help guide all future design decisions. A diverse array of images, textures, and colour palettes is collected to provide visual references for those keywords. Organizing these visual elements into a structured board, digital or physical, serves to convey the overall feeling of a project. Mood boards are used by the designers to communicate subjective aspects of the product to clients in the early phases of the design development (Milton & Rodgers, 2013).

3.4.3 CAD modelling

3D Computer-Aided Design (CAD) models allow to represent the product as a construction of geometric shapes and solid entities. This has the advantage of visualising the 3D forms of the design and to later create realistic rendering of the product to check its appearance. CAD models also let the designer compute mechanical and physical values such as mass and volume properties, as well as to create 2D drawings that can be used for production. This method facilitates rapid modifications throughout the iterative design process and reduces the need for excessive physical prototypes during the early stages of the development (Ulrich & Eppinger, 2016).

3.5 Deliver

3.5.1 User survey

User surveys are used to collect data from a large group of participants regarding their experiences, opinions, behaviours, or preferences. They allow to notice patterns across users because, when compared to interviews, their structured format makes it easier to compare responses. One main advantage of this method are that online surveys can compile the data automatically and can reach a large number of people. Online surveys can help validate design decisions and allow researchers to analyse responses statistically while still collecting additional personal feedback in open-ended questions (Rogers et al., 2023).

3.5.2 Prototyping

Prototypes can be physical, analytical or a combination of both. They facilitate communication with a three-dimensional representation instead of a simple explanation or sketch of the object. Prototypes make sure that the components of the product work well together and that the proportions are right. They can be of low or high-fidelity and can focus on the shape and aesthetics, the functionality, or both in later stages, to allow for rapid improvements before production. Prototyping is a valuable design methodology as they transform ideas into tangible representations, which can be tested and evaluated by users and stakeholders to improve the quality of the final product (Ulrich & Eppinger, 2016).

4 Implementation and Results

This chapter applies the methods discussed in the previous chapter and details the results obtained from it.

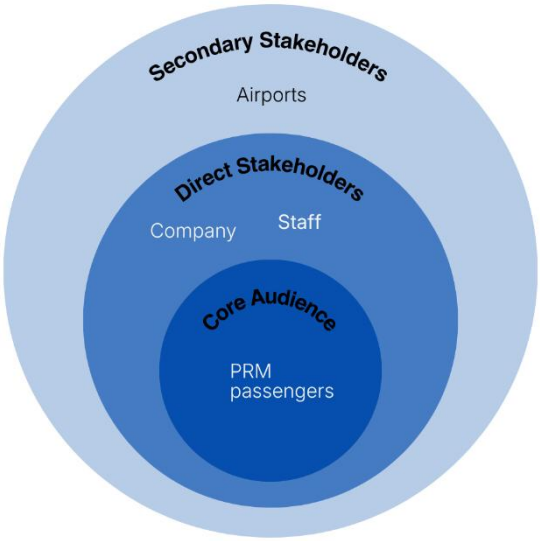
The project followed a human-centred design approach by involving users and stakeholders throughout the design process. Observations, semi-structured interviews, and other methods were used to identify user needs, which informed the development of design requirements and subsequent concept generation. A user survey was later conducted to validate the designs and propose improvements for further research.

4.1 Discover

4.1.1 Stakeholders

In order to prioritise certain functions or design decisions, the stakeholders of the project and their influences were chosen and analysed. Stakeholders for this project were considered as the people who have firsthand contact with the wheelchair, as well as indirect influence on the design. As seen in Figure 2, an onion diagram demonstrates the order of importance of each stakeholder, increasing in significance as they get closer to the centre. The people who will have direct contact and who will use the product the most were the users, so they have the most influence on design decisions, which placed them at the core of the diagram. Next, DAAV's technical team also has a strong influence in the decisions made, as they may request specific features to include for the wheelchair's autonomous navigation. Since the wheelchair may need maintenance or changing the batteries at the end of the day, the staff in charge will interact with the wheelchair as well, therefore some features for the wheelchair will be designed with staff members in mind. As the wheelchair will operate in airport environments, it is also necessary to include airports as a stakeholder. For instance, the design style of airports will have an influence on the wheelchair's style, because the wheelchair must be appropriate for its environment.

Figure 2: *Stakeholders*



4.1.2 Competitor analysis

To guide the design process several market analyses were conducted. Three main areas of focus were chosen: autonomous wheelchairs used in airport environments, general autonomous wheelchairs and electric wheelchairs.

Firstly, the analysis focused on the current competitor market for autonomous wheelchairs specifically used in airport environments (Figure 3). To find autonomous wheelchairs made for airports, those terms were researched using a search engine and the companies that produced this specific product appeared. DAAV also provided leaflets with emerging autonomous wheelchairs for airports that were presented at a recent mobility event. The companies' websites were read and specifications about these wheelchairs were examined, specifically the features tailored to airports were observed. In Figure 4, these autonomous wheelchairs for airports were analysed on their design standpoint alone and from a user's and design perspective. The "+" sign represents a positive feature, while the "-" sign represents a negative feature. Collecting these insights was useful to understand what features seemed to have a positive effect and could be used or further developed.

Figure 3: *Autonomous airport wheelchairs*



Figure 4: *Competitor analysis of autonomous wheelchairs for airports*

	+ User	- User	+ Design	- Design
ALBA Robot	Light ring on floor Videocall assistance	Flat, metal armrest Seat seems high High lugagge carrier	Mascot&meaning Stable pole Light details	Bulky style Edges Lidar pillars Not very dicreet
Blueberry BBGo	Low carrier Manual push Side access Seat detection	Not welcoming Small sensors	Visibility pole light Airport theme	Mechanical design No omniwheels Not very welcoming
WHILL	Seat padding Stable footrest Manual push	Carrier is a bit tight and high Makes noise and sounds often	Cohesive design Light details Omniwheels Good materials	Looks a bit bulky or heavy
i-MOBBY	Accessible emergency stop Seat looks comfortable	No side access Small sensors	Cohesive More design oriented than mechanical Stackable	Huge base No changeable seat cushion
LIBPET	Feels protected Large seat	Legs are blocked	Friendly colors	Forklift look

It is relevant to note that all autonomous wheelchairs have sensors, as they are necessary components to be able to navigate autonomously. It is common to combine both exteroceptive sensors, which can perceive the external environments, and proprioceptive sensors, which monitor the device's internal state. It was observed that for airport and general autonomous wheelchairs Light Detection and Ranging (LiDAR) and cameras were used as exteroceptive sensors and some also used proprioceptive sensors such as Inertial Measurement Units (IMUs). LiDARs allow to 3D map the environment and to conduct obstacle detection in real-time. Using 2D and 3D LiDAR sensors simultaneously can facilitate detection of smaller objects and steps. IMUs are comprised of accelerometers, gyroscopes, and magnetometers that measure acceleration, angular velocity, and orientation. Combining the data collected from the exteroceptive and proprioceptive sensors helps overcome the limitations of any single sensor and ensures high reliability for autonomous navigation. For airport autonomous wheelchairs, the 3D LiDAR sensors could be seen placed high above the wheelchair or lower to the ground, such as on the armrests.

Their luggage compartments were similarly examined. They were mostly of squared shapes and often placed on the backrest of the chair, which could increase the difficulty and strain when placing the luggage inside.

Secondly, general autonomous wheelchairs served as inspiration for different sensor or screen placements, as well as overall form analysis. These wheelchairs were found from sites such as Pinterest, Google, Yanko Design and Red Dot Design Award. All kinds of autonomous wheelchair shapes and types were chosen and they were narrowed down to nine different wheelchairs, which was deemed enough to analyse and compare features (Figure 5). It was noted that a few of the wheelchairs installed their sensors on top of the two front wheels, on the armrests or on the base, whereas, for the airport wheelchairs, 2D sensors could also be hidden under the base.

Figure 5: *Autonomous and electric wheelchairs*



Additionally, electric wheelchair designs and their battery placements were analysed.

Common colours in motorized wheelchairs often included dark colours for the seat pads, such as black and dark grey. This is likely because the seat can easily get dirty and darker colours hide stains better. For the frame, white was the most common and black or grey were also used. All motorized wheelchairs had four wheels, and, in some cases, the back motorized wheels were larger than the front wheels.

Three main types of wheelchair designs emerged in how to assemble the wheels, base, and seat. The first consisted of a thin base connecting all four wheels, with an inclined support extending up to the seat. The second used a thicker base and incorporated a large block directly beneath the seat. The third did not have a continuous bottom platform, instead it linked each pair of wheels directly to the seat base through separate supports.

4.2 Define

4.2.1 Persona

To support the development of a user centred design proposal, a persona was created to empathise with the users and to better understand their needs and behaviours. The principal user identified for the autonomous wheelchair is François Dunand, a 79-year-old French grandfather, who travels on his own to visit his family during holiday periods. Although he can walk short distances, prolonged walking causes him excessive knee pain, meaning he can find large and complex airport terminals demanding and exhausting to navigate. He usually carries a small suitcase and uses a cane for additional support.

He values independence and dignity and prefers solutions that allow him to remain self-sufficient throughout his journey. François feels uncomfortable using products that appear overly medical, as they can make him feel stigmatised and different from others.

While he is capable of using basic technology, François prefers interactions that are simple, intuitive, and easy to understand. Complex interfaces or unclear instructions can quickly become stressful, especially in unfamiliar environments such as airports. Therefore, the autonomous wheelchair must provide clear guidance, accessible controls, and a reassuring user experience that supports confidence and ease of use during travel.

4.2.2 Try it yourself

Two autonomous wheelchairs for airport use, developed by DAAV, the AIR (Figure 6) and the B.AIR (Figure 7), were tested and analysed. The designer acted as the user and followed a user journey visualised in Figure 8. Each wheelchair and their respective features were tested.

Figure 6: *DAAV AIR*



Figure 7: *DAAV B.AIR*



It is important to recall that the "Try it yourself" method (3.3.2) can support empathetic product design, as the designer temporarily takes on the role of the user while testing the product. This approach enables the rapid identification of several issues and user needs. However, because there is a risk of personal bias, the method was complemented with semi-structured interviews and observations.

Figure 8: *User journey visualisation*

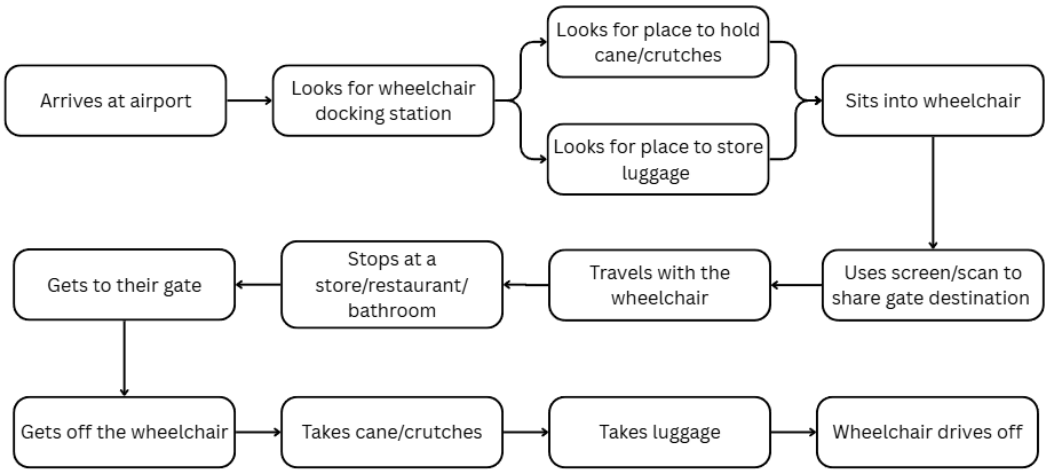


Figure 9 shows a few pictures taken during the testing of both wheelchairs. Following the first stages of the user journey, it was noted that the wheelchairs' tall LiDAR poles made them easier to spot and identify from a distance. Secondly, the designer's previous experience with the B.AIR wheelchair (1.2) concluded that placing the compartment lower to the ground allowed users to lift their belongings to a lower height compared to that of competitors and for a shorter period of time. Consequently, users experienced less physical strain.

Figure 9: *Try it yourself pictures*



Furthermore, after storing the luggage, users could choose to leave their cane or crutches before sitting down. The AIR wheelchair did not include a dedicated compartment for this purpose, whereas the B.AIR featured a holder integrated into the side of the luggage box. However, this meant that the users needed to walk around the side of the device before sitting down. One desirable feature of the B.AIR is its liftable armrests, which allowed users to access the seat from the side of the wheelchair and reduce the distance they need to walk to sit down. A thoughtful improvement, that was later considered for the final design, could be not only permitting side entry, but also placing a holder closer to the user so they can continue using their mobility aids until seated and then store the aid afterwards.

Additionally, it was noted that armrests with soft cushioning improved overall comfort and created a more pleasant seating experience. The B.AIR wheelchair included ergonomic qualities such as an angled positioning of the seat and backrest, together with the inclusion of a wide footrest that provided better support and relaxation during use.

4.2.3 Observations

To develop a better understanding of the experiences of passengers with reduced mobility, an observational study was carried out within the assistance area for PRM

passengers at the Geneva Airport over the course of a weekend. By remaining in the assistance area for several hours, it was possible to identify repeated behaviours and issues that emerged between passengers, airport staff, and wheelchairs.

Rather than viewing mobility assistance purely as a functional transport service, the observations suggested that the experience is shaped equally by emotional, social, and practical factors. One recurring issue concerned the extended waiting periods experienced by many passengers before a wheelchair or staff assistance became available. During these periods, users were often required to manage their personal belongings while remaining seated in the area or on a wheelchair. Because the existing wheelchairs provided no dedicated storage, bags were commonly balanced on passengers' laps, placed between the legs, or positioned directly on the footrest. Backpacks were occasionally hung behind the push handles, although this solution appeared improvised rather than intentional. Other items such as coats, shopping bags, or handbags frequently had to be held throughout the waiting process. While this situation may initially appear minor, it contributes to reduced physical comfort and a diminished sense of dignity.

The observations also dismissed the common assumption that airport wheelchair users primarily consist of individuals with permanent disabilities. It was shown that a considerable proportion of passengers appeared to be temporary users recovering from injuries. Knee braces, arm braces and crutches were often observed throughout the study period. These users also might not be familiar with using manual wheelchairs themselves. This finding reflects the challenges around autonomy and dependence when it comes to mobility assistance services. Although some passengers required physical transport assistance, others appeared to only need the wheelchair for easier mobility.

In one instance, a young passenger using crutches and a leg brace became visibly frustrated after being informed that wheelchair access within the PRM service required mandatory staff accompaniment. The passenger repeatedly expressed that he did not want an escort and only wished to use the wheelchair independently. This interaction revealed a tension that is frequently discussed within disability and assistive design: solutions intended to support users may still reduce their sense of dignity when autonomy is restricted through mandatory procedures. Independence, therefore, cannot be understood through physical capability alone. It is also shaped by how services permit users to maintain control over their movement, decisions, and social visibility within public spaces.

Elderly passengers were frequently observed carrying canes while seated, but with no storage options to secure them safely during transport, users often had to balance their mobility aids themselves while simultaneously handling personal belongings. Collectively, these findings indicate that current airport assistance systems tend to prioritise transportation efficiency and operational procedure over broader experiential concerns. Issues such as temporary injury accommodation, storage accessibility, emotional reassurance, and perceived independence appeared underdeveloped despite their evident influence on the overall travel experience.

4.2.4 Semi-structured interviews

To complement the observational study, a series of semi-structured interviews were also conducted within the assistance area for PRM passengers in Geneva Airport. This method was selected in order to capture perspectives that could not be fully understood through observation alone, particularly subjective experiences and sensitive topics relating to comfort, independence, dignity, and safety. Although observations showed how passengers interacted with the existing service, semi-structured interviews made it possible to explore their emotions and expectations. A more flexible interview structure, known as a semi-structured interview, was therefore used. Several questions were prepared beforehand and are included in 7.1 Appendix A: Passengers semi-structured interviews with their answers. However, being open to conversations and natural deviations allowed participants to elaborate on their opinions.

Throughout the research, ethical standards followed the guidelines of the Human Research Act (HRA), where participation must be informed and voluntary, and where privacy and data security are essential (Federal Office of Public Health [FOPH], n.d.). Interviewees were adults capable of making independent decisions, so they were not considered vulnerable individuals requiring additional protective measures. Before each interview, participants were orally informed of the purpose of the research, the anonymous use of their answers for academic purposes, as well as their right to withdraw at any time since participation was voluntary. Confidentiality was ensured by anonymising all collected data and excluding personal identifiable information from the analysis and documentation. Therefore, approval from the HRA was not required and oral informed consent was considered sufficient and appropriate for this research context.

Passengers using the PRM service were approached directly in the waiting area and invited to participate in an interview to share their experiences with airport mobility assistance. The thirteen anonymised semi-structured interviews included elderly travellers, passengers with temporary injuries, and visually impaired users, ranging from their mid-thirties to late eighties, reflecting the broad spectrum of people relying on mobility support within airport environments. Additionally, an anonymised semi-structured group interview, detailed in 7.2 Appendix B: Staff semi-structured interviews, was held with six members of the airport assistance staff, aged from their twenties to late thirties, in order to understand the operational challenges associated with transporting passengers and managing wheelchairs, as well as to discover any recurring issues.

One recurring topic concerned the practical limitations of the wheelchairs currently provided at the airport. Across many discussions, users identified the absence of dedicated storage as a persistent issue. Luggage, purses, crutches, and canes were frequently mentioned as items that had to be managed while seated. Several participants described keeping belongings on their lap or positioning them near their feet during transport; a behaviour that stood out during the observation study. One passenger who had recently injured his knee reacted humorously but pointedly to the question of luggage storage, remarking: “I broke my knee so I can’t put my bag on my lap now can I?” Although informal, comments such as this revealed how design details significantly influence comfort and usability during travel. Staff members reinforced these concerns from an operational perspective, repeatedly emphasising the difficulties involved in carrying luggage while simultaneously pushing one or multiple passengers through crowded airport spaces.

The semi-structured interviews also suggested that airport wheelchair users had different expectations regarding assistance. On one hand, participants with temporary injuries often expressed a stronger desire for independence and viewed the wheelchair primarily as a practical mobility tool rather than a source of personal support. On the other hand, some elderly or visually impaired users associated the PRM service with reassurance and human interaction. One visually impaired participant explained that she preferred assistance from staff because she relied on verbal communication and guidance throughout the journey, which can be taken as a valuable design feature for the company to implement in the autonomous wheelchair's software.

Opinions toward the concept of an autonomous wheelchair were varied. Some interviewees believed that a self-driving wheelchair could strengthen independence by reducing reliance on staff. This view was common among passengers with short term injuries who did not necessarily require continuous staff assistance. Others, however, expressed concerns regarding collision avoidance, navigation reliability, and communication. Visually impaired users, for instance, insisted on the importance of auditory feedback and reassurance throughout the journey. One participant explained that she would want the wheelchair to continuously communicate where it was going and whether it was taking the correct route. Another participant emphasised that unnecessary warning sounds could become overwhelming due to sound sensitivity. Such comments illustrate how accessibility in autonomous devices cannot be reduced solely to physical functionalities. Sensory experience, communication methods, and emotional reassurance appear equally important within inclusive mobility design. These user needs are essential and need to be taken into account. However, as a reminder, since this thesis focuses on the design of the hardware side of an autonomous wheelchair, the user needs will be collected with that limitation in mind.

Moreover, questions relating to appearance and aesthetics of a possible autonomous wheelchair gave interesting responses. The users were given three possibilities to choose from: friendly and reassuring (Figure 10), medical and safe (Figure 11) or modern and innovative (Figure 12).

Figure 10: *Friendly and reassuring design style*

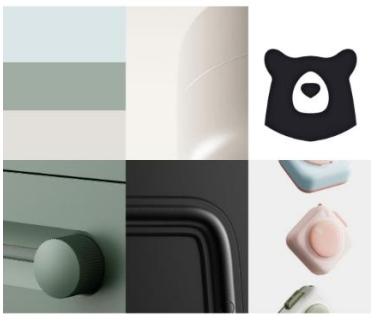


Figure 11: *Medical and safe design style*



Figure 12: *Modern and innovative design style*



Only one elderly user preferred a more medical appearance because it felt familiar and trustworthy, while others associated medical aesthetics with stigma, ageing, or social discomfort. Two participants chose the friendly appearance, while most other users, generally younger ones, were leaning more towards a modern and innovative design language. These results show that although inclusive products should maintain a high focus on functionality, their aesthetic appearance does influence how users feel when interacting with them. Although, it is interesting to note that not all interviewees considered aesthetics important, as two participants mentioned that reliability and functionality were more important than the visual appearance of the wheelchair.

The interview with staff members showed another insight into the logistical aspects that shape the overall mobility experience. Employees described difficulties pushing multiple wheelchairs at once, transporting luggage, and passing narrow automatic doors with two wheelchairs. Staff members noted that wheelchairs deteriorated under heavy use, which created difficulties in manoeuvrability and could also harm the passenger. They also often found themselves in a rush and had to push multiple wheelchairs at once or rush a passenger to their gate. These answers validate the need to provide extra mobility assistance to lessen the strain on staff members and to improve PRM's safety.

Taken together, the semi-structured interviews revealed that airport mobility assistance involves a complex interaction between physical support, emotional reassurance, operational efficiency, and personal autonomy. Although participants generally appreciated the existing PRM service, recurring frustrations surrounding waiting times, luggage handling, and independence indicate opportunities for improvement. Rather than replacing human assistance entirely, the findings suggest the value of a joint system between an autonomous wheelchair and staff assistance, which manages to accommodate different levels of independence according to the needs and preferences of individual users.

4.2.5 Design requirements

To identify the design requirements and constraints, user needs were first collected and classified. The classification of user needs was carried out through a thematic analysis, in which recurring patterns across the previous methods (try it yourself, observations, and semi-structured interviews) were identified and grouped into broader themes and subthemes. The analysis distinguished between needs that appeared more frequently (labelled High to Low Frequency), as well as those of the company itself (labelled Company). Therefore, frequency was used as an important indication for prioritisation. This shows that product requirements are often shaped not only by end-user expectations, but also by logistical constraints and those of other stakeholders.

The resulting prioritisation, shown in Table 1 below, follows a structure in which "Must-have" requirements represent frequently requested functions from the discussions and observations, which are considered essential for the wheelchair to operate safely, comfortably, and effectively within the intended airport setting. "Should-have" and "Could-have" categories include features mentioned less frequently, that may improve usability or user confidence but were considered less critical to the core objectives of the project. Some of them will nevertheless be considered in the concept generation phase. By contrast, software-focused needs were intentionally classified as "Won't-have" requirements, not because they lack importance in autonomous mobility systems, but because they fall outside the scope of this thesis, which focuses specifically on the physical design of an autonomous wheelchair for airports.

Table 1: *Classification of user data collected through previous methods*

Theme	Subtheme	User Data	Type	Frequency	Priority
Physical Comfort & Ergonomics	Seating comfort	Sit comfortably during extended airport use	Hardware	High	Must-have
Physical Comfort & Ergonomics	Lower body support	Rest feet safely while seated	Hardware	High	Must-have
Physical Comfort & Ergonomics	Upper body support	Rest arms while using the wheelchair	Hardware	High	Must-have

Safety & Reliability	Weight support	Support users of different body weights safely	Hardware	High	Must-have
Independence & Autonomy	Independent transport	Reduce dependence on airport staff for mobility assistance	Soft+Hardware	High	Must-have
Storage & Belongings	Personal belongings	Store luggage and personal belongings	Hardware	High	Must-have
Independence & Autonomy	Mobility aid integration	Store canes or crutches during transport	Hardware	High	Must-have
Navigation & Information	Destination input	Enter or communicate gate destination easily	Soft+Hardware	High	Must-have
Accessibility & Inclusive Use	Tactile accessibility	Distinguish controls through tactile interaction	Hardware	High	Must-have
Physical Comfort & Ergonomics	Transfer accessibility	Access and exit the wheelchair with minimal strain	Hardware	High	Must-have
Operational & Maintenance Needs	Battery management	Recharge or replace batteries efficiently	Hardware	Company	Must-have
Operational & Maintenance Needs	Storage efficiency	Store multiple wheelchairs compactly	Hardware	Company	Must-have
Operational & Maintenance Needs	Maintenance accessibility	Access internal components for maintenance easily	Hardware	Company	Must-have
Operational Needs	Autonomous navigation	Navigate safely within dynamic airport environments	Hardware	Company	Must-have
Safety & Reliability	User reassurance and trust	Feel confident and safe while using the wheelchair	Soft+Hardware	Medium	Should-have
Operational & Maintenance Needs	Manual override	Allow manual movement if power is unavailable	Hardware	Medium	Must-have
Storage & Belongings	Secure luggage transport	Keep luggage visible or protected during travel	Hardware	Low	Could-have
Physical Comfort & Ergonomics	Extended leg accommodation	Support users with extended leg braces	Hardware	Low	Could-have
Navigation & Information	Route awareness	Be informed about wheelchair route and destination	Software	High	Won't-have
Accessibility & Inclusive Use	Accompanying travel support	Allow companions to travel comfortably alongside users	Software	Medium	Won't-have
Navigation & Information	Multilingual communication	Support communication in multiple languages	Software	Medium	Won't-have

Accessibility & Inclusive Use	Audio accessibility	Provide audio guidance for visually impaired users	Software	Medium	Won't-have
Independence & Autonomy	Autonomous pickup	Request wheelchair pickup from designated airport areas	Software	Low	Won't-have
Accessibility & Inclusive Use	Emotional support	Maintain opportunities for human interaction and assistance	Software	Low	Won't-have
Navigation & Information	Travel information updates	Receive gate and route updates during travel	Software	Medium	Won't-have
Safety & Reliability	Collision avoidance	Avoid collisions with people and obstacles	Software	High	Won't-have
Safety & Reliability	Controlled movement	Maintain safe movement speeds in airport environments	Software	Medium	Won't-have
Safety & Reliability	User control over autonomy	Allow users to modify or cancel destinations	Software	Medium	Won't-have

Following this classification stage, the identified needs were translated into design requirements that can directly inform the physical development of the wheelchair. The transition from user need to design requirement involved reformulating qualitative insights into actionable product characteristics. For example, as seen in Table 2, the recurring need for users to store personal belongings was translated into an integrated storage solution. Similarly, requirements provided by the company itself, including compact wheelchair storage or accessible battery maintenance, influenced structural layout and component organisation of the wheelchair.

Table 2: *Design requirements set from primary user needs*

User data	User need	Design requirements
Sit comfortably during extended airport use	The wheelchair should support comfortable seated postures during prolonged use	Ergonomic seat and backrest
Rest feet safely while seated	The wheelchair should provide stable lower-body support during movement	Integrated footrest
Rest arms while using the wheelchair	The wheelchair should provide upper-body support during use	Armrests
Support users of different body weights safely	The wheelchair should safely support users weighing up to 120 kg	Strong structural base
Reduce dependence on airport staff for mobility assistance	The wheelchair should support autonomous movement within airport environments	Autonomous navigation system
Store luggage and personal belongings	The wheelchair should allow users to transport personal belongings independently	Integrated luggage compartment

Store canes or crutches during transport	The wheelchair should accommodate common mobility aids securely	Mobility aid holder
Enter or communicate gate destination easily	The wheelchair should support simple and accessible destination input	Scanner, touchscreen, or camera interface
Distinguish controls through tactile interaction	The wheelchair controls should support tactile and visual accessibility	Haptic buttons
Access and exit the wheelchair with minimal strain	The wheelchair should support accessible transfers for users with reduced mobility	Accessible sides, movable armrests and footrests
Recharge or replace batteries efficiently	The wheelchair should allow efficient battery charging and replacement	Removable or swappable battery
Store multiple wheelchairs compactly	The wheelchair should minimise storage space requirements	Stacking or nesting mechanism
Access internal components for maintenance easily	The wheelchair should provide safe access to electronic and mechanical components	Accessible maintenance compartment
Navigate safely within dynamic airport environments	The wheelchair should detect obstacles and navigate safely in crowded environments	LiDAR and sensor system
Feel confident and safe while using the wheelchair	The wheelchair should communicate safety and reliability through its appearance and interaction	Reassuring form language and interface
Allow manual movement if power is unavailable	The wheelchair should support manual handling during system failure	Push handle

Although the main focus remained on “Must-have” requirements, selected “Should-have” features were also taken into account for the design where they meaningfully contributed to usability and user experience. This approach balanced user-centred design with practical constraints, demonstrating that inclusive mobility devices in complex public environments must consider simultaneously end-user expectations and technical considerations.

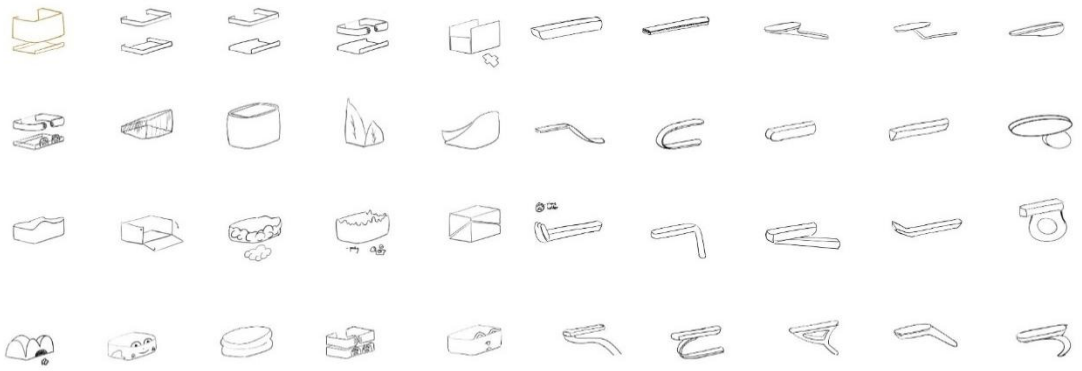
4.3 Develop

4.3.1 Brainstorming

A couple of brainstorming sessions were held throughout the Develop phase to generate ideas for the design of the autonomous wheelchair. The ideations involved sketching sessions that allowed to come up with concepts without restrictions on the first iterations. This method was used for the designer to keep an open mind and a large selection of starting points for the design direction that will be narrowed down in the concept generation phase.

The first sketching sessions focused on separate wheelchair components. Figure 13 shows a collection of luggage compartments and armrests. The ideas proposed were a combination of friendly designs, that included the use of animal and natural elements to create a more expressive design, and more modern shapes with simple curves and sleek lines.

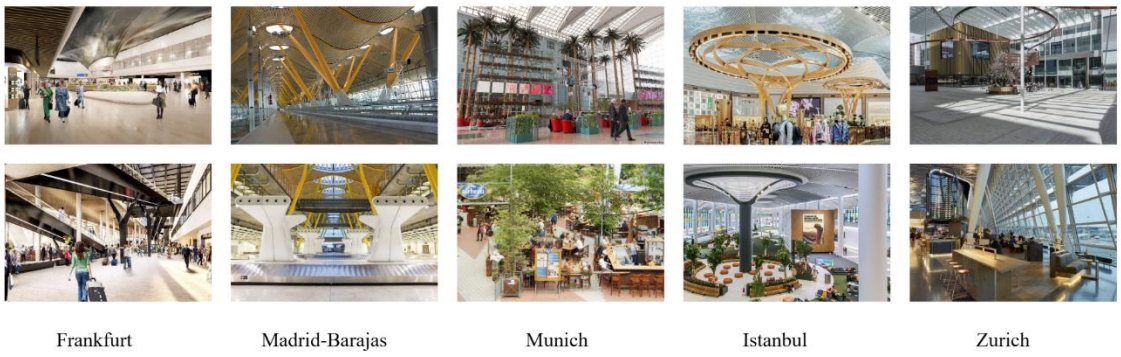
Figure 13: *Luggage compartment and armrest sketches*



4.3.2 Airport context

Five European airports, shown in Figure 14, were analysed to better understand the environments that the autonomous wheelchair would be used in and to align the design direction in a compatible manner. The selected airports included Frankfurt Airport, Adolfo Suárez Madrid-Barajas Airport, Istanbul Airport, Munich Airport, and Zurich Airport. The analysis focused on architectural details such as forms, material choices, and colours, as well as the general atmosphere produced within these spaces.

Figure 14: *Selection of pictures from airport analysis*



Several recurring features were observed. Tall glass walls, large circulation areas, and elevated ceilings were used to maximise natural light and reinforce a sense of openness. Other structural elements, such as exposed beams and metallic structures, created the impression of a robust design. At the same time, many airports appeared to soften their

mechanical look through the incorporation of organic shapes and natural materials. For instance, curved decorative ceilings, rounded arches, indoor vegetation, and wooden details were repeatedly observed. These choices seem to come from a biophilic design approach, which argues that the integration of natural elements within an environment can positively affect people's well-being (Kellert, 2018). In large stressful environments, where passengers often experience fatigue or pressure, softer shapes and natural materials, or also a natural colour palette, can improve passenger's well-being and help shape the overall user experience.

Additionally, the ways in which colour was presented and used in certain terminals was observed. The most dominant colours appeared to be neutral tones, with white, grey, beige, black, and natural wood finishes being used the most for floors, walls, and large decorative structures. Some brighter colours, such as red, blue, green, and yellow were present, but found mostly in certain furniture. These pops of colour introduced variation without overwhelming the environment. The overall palette appeared to support a calmer and more cohesive atmosphere while still allowing slight visual contrast. Furthermore, several airports incorporated more technology driven elements in premium areas. Glossy black finishes and integrated lighting were commonly used to communicate a luxurious and modern feeling.

This airport environment analysis reinforced the idea that the wheelchair's design should not be overly medical but instead fit more naturally within the airports' architecture and atmosphere. It should therefore include soft transitions, organic shapes and mostly neutral colours. Discreet lighting integration can add a modern element to the wheelchair. By aligning the wheelchair with the spatial and material qualities already present in airport environments, the visual separation often experienced between assistive devices and the public spaces in which they operate is reduced and leads to an increase in users' well-being.

4.3.3 Keywords and mood board

To funnel down the design direction of the project, keywords were generated based on the user needs and previous research to define how the autonomous wheelchair should make the user feel and how it should look. This led to establish a few design consequences, described in Table 3, to satisfy that intent.

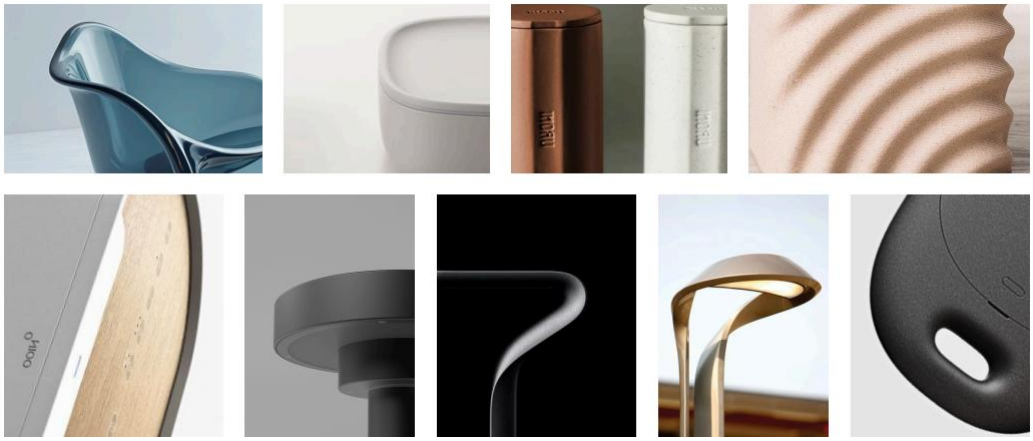
Table 3: *Synthesis of keywords and their corresponding design consequence*

		Design consequence
Feel	Safe	Balanced chair, robust materials, modern technology
	Confident	Not too flashy design, not overly medical/technical
	Calm	Use of slow curves, organic shapes, natural materials
Look	Friendly	A bit of colour, rounded shapes, slow curves, chair has character/personality (allusion to animals, people, nature)
	Cohesive	All elements fit together, in the same design style
	Modern	Screen displays, light integration, sleek design
	Comfortable	Cushions, ergonomic posture
	Intuitive	Easy to use, few buttons, clear functions

Since plastic made up a large portion of the wheelchair's materials, the thought of trying to use natural or sustainable materials resulted in choosing to use recycled ocean plastics when possible, for example for the black base or the outer shells of some parts mentioned in the next chapters. Reinforcing recycled polypropylene with glass fibres would keep the parts quite stiff and allow for a more sustainable product. The combination of ocean plastic, organic shapes and a calm feeling brought about a central design proposal: waves. The autonomous wheelchair would be inspired by a wave. It reflects movement and emotion, as it carries users toward their destination while creating a sense of calm and comfort. Using soft, organic shapes would create an approachable presence, transforming the wheelchair from a clinical device into a supportive travel companion. This follows the first principle of universal design (UD) for equitable use, which aims to create designs that avoid stigmatising any users and are appealing to all users (CEUD, n.d.).

To complement the previous keywords and inspirations, a mood board (Figure 15) was created. In it slow curves, recycled material, and neutral colours with a pop of blue are shown. The mood board also reflected a modern feel, with matt black textures and simple but elevated shapes, as well as integrated lighting. The wave element appeared in the pop of blue colour and the large curves, but it was also present in the use of recycled ocean plastic parts. In order to design an intuitive mobility device, simple surfaces and interfaces had also been selected to showcase the ease of use of the product.

Figure 15: *Mood board for the design of an autonomous wheelchair for airports*



4.3.4 Concept generation and selection

The concept generation phase stems from the continuation of the brainstorming sessions and the parallelly found design directions in the previous chapters.

i. Wheelchair ideation

A sketching session was organised focusing on entire wheelchair models. During this session multiple wheelchair models were drawn based on the first concepts previously

shown in Figure 13, and new ideas were explored by following previous shapes or structures. This allowed to play around with the shape of the seat and backrest, as well as to experiment with the base. These elements would be placed in a way that would create a consistent aesthetic design. Figure 16 still shows some slight experimentation with the luggage compartment's shape, such as an open curve that follows the backrest or a basket supported by a V shape. These rounded solutions were considered an interesting concept that has not been seen in competitor products. However, a squarer shape was chosen because it can hold the luggage in a safer manner while also reducing material use.

The research on airport contexts and the mood board was conducted in parallel to some of the wheelchair sketching sessions. This can be seen at the bottom of Figure 16, where seashell elements and curvy backrests started to be iterated.

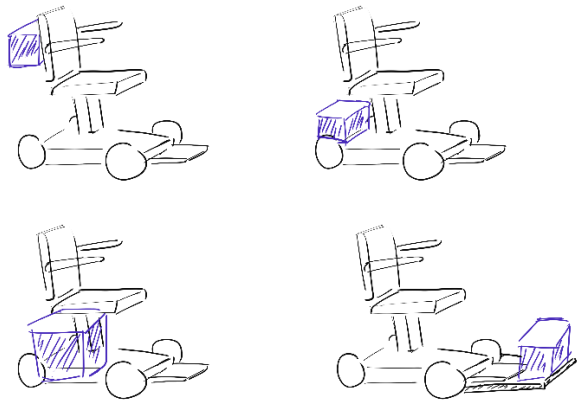
Figure 16: *Wheelchair sketching*



The placement of the luggage compartment was chosen at the back of the wheelchair's base for several reasons. Figure 17 shows a few positions that were considered: on the backrest, on either side, at the front or at the back of the base. A visually impaired interviewee had mentioned wanting to have their belongings close to them (Appendix A: Passengers semi-structured interviews), so the concept of attaching the compartment to the side of the wheelchair would satisfy that. Nevertheless, it created an instability and reduced the safety of the device, as well as blocked the users from sitting down on one side of the wheelchair, making it less accessible. Placing the luggage holder at the front increased the length of the wheelchair significantly because there would need to be enough space for the users' legs first before being able to place the compartment. Both concepts at the back of the wheelchair seemed the most practical, but having the compartment attached to the backrest, similarly to many competitors, increases the difficulty for PRM passengers to be independent. Indeed, it means that they have to lift up their luggage quite high before managing to place it into the holder. Therefore, placing it as low as possible to the ground reduces the physical strain imposed on the users and permits them to effectively use the autonomous wheelchair. This last concept, as chosen with the B.AIR wheelchair (4.2.2), was considered the most inclusive and

accessible design choice, allowing the users to maintain as much independence as possible.

Figure 17: *Luggage compartment position ideation*



Later, the AI tool Gemini was used to generate and test visuals of various concepts of wheelchairs stemming from the theme of ocean and waves. The following prompt generated the wheelchairs seen in Figure 18:

"Generate an electric wheelchair that has a modern and minimalistic design. It has four wheels of the same size, two armrests. the space at the back of the wheelchair to fit a luggage, it is square and should be able to fit a hand size luggage, it has organic details inspired by a seashell shape. The whole colour palette includes dark blue, off white, grey and black. The design is intuitive, calm, modern."

Figure 18: *AI generated visualisations of the "wave" design direction*



Although the instructions were not precise enough to generate an accurate representation of what was wanted, this step confirmed that the previous design choices were going in the desired direction.

The colour palette is pleasant: the white and cream colours are elegant, while the blue keeps a calm and approachable atmosphere. Adding a touch of vibrant colour also lessens the medical appearance that a usual wheelchair would have. Incorporating slower curves in the design adds to the comfortable and calm design language and minimising the number of buttons and functions keeps the wheelchair intuitive.

ii. Wheel type and placement

One goal for the project was to design an autonomous wheelchair capable of omnidirectional or holonomic motion. Holonomic systems have been defined as platforms where the degrees of freedom that can be controlled are the same as the total degrees of freedom. This implies that such a device can independently translate and rotate. Using mecanum or omnidirectional wheels (Figure 19), which feature rollers oriented at a certain angle, can allow for lateral movement without turning the chassis.

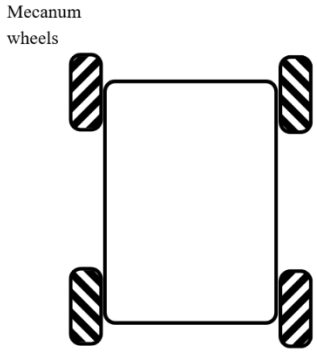
Figure 19: *Type of wheels considered for an omnidirectional or holonomic movement*



Mecanum wheels Omnidirectional wheels

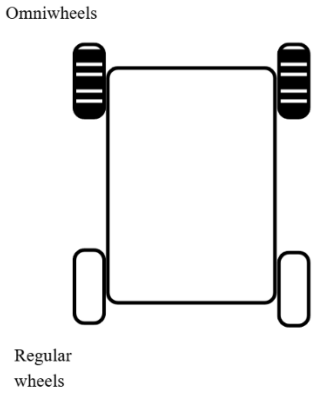
An initial concept considered the use of four mecanum wheels. In this configuration, the wheels are aligned with the chassis, while their rollers are oriented at approximately 45° (Figure 20), enabling the generation of lateral forces. When driven in coordination, this arrangement can produce holonomic motion, allowing the wheelchair to translate in any direction and rotate about its vertical axis without reorientation. Such systems are common in mobile robotics and offer a high degree of manoeuvrability. However, this approach introduces notable drawbacks. The use of four mecanum wheels increases both mechanical complexity and cost and may also reduce energy efficiency. In large environments such as airports, where spaces to circulate and turn are generally unobstructed, aiming for full holonomic capability may not be worth these added constraints.

Figure 20: *Mecanum wheel placement*



A mixed wheel arrangement with two front omnidirectional wheels and two standard rear wheels, all of which are in line with the wheelchair base, is a more practical compromise (Figure 21). Although it does not offer true holonomic motion or sideways translation, when this composition is compared to traditional caster wheels, the fixed omnidirectional wheels are more compact because they don't need to turn or reorient while in motion, which means it is also safer for the users sitting on the wheelchair, as their clothes or feet can't get caught in the wheels. Having two regular rear wheels also helps lower the costs and the omnidirectional front wheels allow for a small turning radius.

Figure 21: *Omni wheel and regular wheel placement*



iii. Wheelchair storage solutions

The next step was to design the wheelchair in a way that could reduce the space taken up by the devices when not in use. This would enable airports to store the devices efficiently and additionally reduce storage cost. A couple of solutions were considered for this matter.

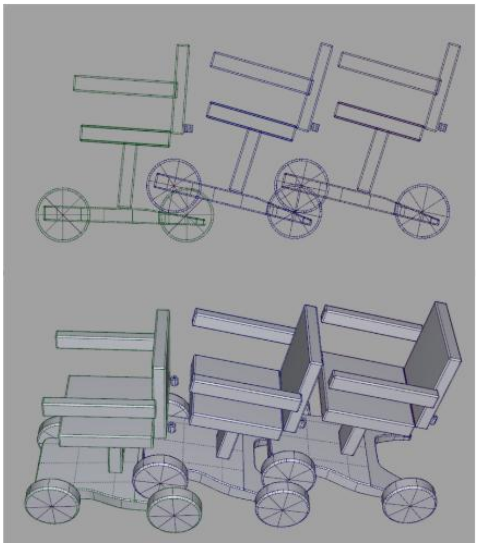
A stackable concept would mean that the wheelchairs would be able to pile on top of each other vertically. However, this would likely necessitate an external structure that would hold the weight of each wheelchair placed on top of another, in a similar manner

to a vertical car parking. This concept would require an additional manufacturing cost and would need the LiDAR poles to be manually removed before stacking, as they could compromise efficient stackability and could break.

Therefore, nesting, a solution to reduce storage space in a horizontal way, was preferred. Nesting consists of one wheelchair fitting into the frame of another to create more compact rows.

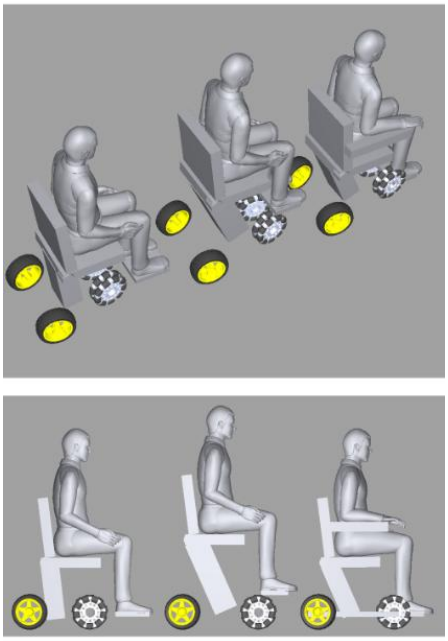
A first solution for partial nesting, observed in Figure 22, was proposed. This concept would allow a wheelchair to roll up the back of the one in front of it, thus saving storage space. Although the LiDAR pole would not need to be unmounted, the luggage holder would need to be manually removed. Moreover, the base would need to be able to support the weight of another wheelchair, and the back wheels would need to be placed significantly wider. This would increase the overall width of the wheelchair which would not allow it to fit into standard European elevators. Elevator dimensions follow the EN 81-70 standard across Europe, which finds the minimum elevator door width to be around 800mm (KONE, n.d.). This means that the maximum width for the wheelchair should be around 700mm to operate safely within elevators' infrastructure.

Figure 22: *First nesting solution*



A solution to keep the width of the wheelchair under 700mm and allow the wheelchairs to fully nest, was to have narrow front wheels. Nevertheless, having narrow front wheels caused a problem regarding the placement of the footrest. As seen in Figure 23, the front wheels would need to be spaced very narrowly to fit in between the back wheels, in order for the back wheels to not be extremely far apart. In this case, the front wheels would interfere with the positioning of the footrest in two ways: either the footrest would need to be placed on top of the front wheels, or the front wheels would need to move backwards for the user to place their feet comfortably. The first concept meant that the height of the wheelchair increased dramatically and it would become unsafe for elderly or injured passengers to use the device, whereas the second concept led to an unstable wheelchair because the front and back wheels were too close together.

Figure 23: *Footrest placement for the second nesting solution*



The final solution was to design a base and the elements on it, such as the footrest and luggage compartment, to allow partial nesting. This meant that instead of the wheelchairs fully nesting into each other, similarly to shopping carts, they would be stored side to side and partially nested, meaning one back wheel of each wheelchair would fit in between the two front wheels of the previous wheelchair, later shown in Figure 37.

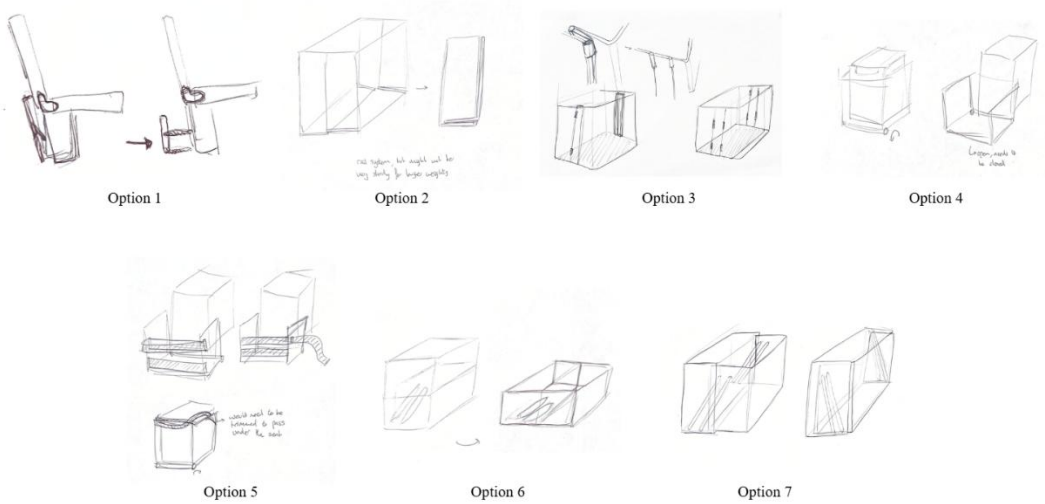
This concept allowed the width of the wheelchair to remain under 700mm and to keep the luggage compartment at the back of the wheelchair's base. Compared with storing two wheelchairs, one in front of the other, in an open configuration (2114 mm), the proposed partially nested configuration (1625 mm) reduces the required storage length by 489 mm, corresponding to a space saving of approximately 23.1%. However, to be able to store the wheelchairs as close as possible, the luggage compartment would need to be adjustable in size, and the footrest would need to fold into the wheelchair.

iv. Luggage compartment mechanism

The luggage compartment's size was designed based on the common dimensions of a cabin bag in Europe. These dimensions are 550 x 400 x 230 mm, as seen with Lufthansa (n.d.). Since it is usual for passengers to overpack their bags and cause them to become larger, it was deemed necessary to add a buffer margin. It is not necessary to make the compartment as high as the cabin luggage, so positioning it slightly higher than the centre should be enough to make sure the luggage stays safely put. Therefore, the chosen internal dimensions were: 260mm wide, 432mm long, and 352mm high.

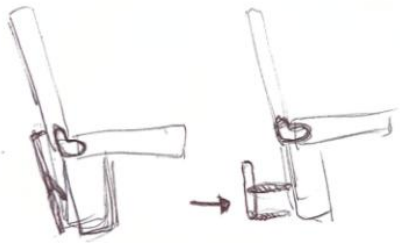
Next, several propositions, seen in Figure 24, for folding the luggage box were considered. It is relevant to note that, for nesting purposes, the base of the wheelchair did not extend past the front or back wheels, which means that the baggage compartment would need to fold far enough so as not to expand past the back wheels when folded.

Figure 24: *Folding mechanisms for the luggage compartment*



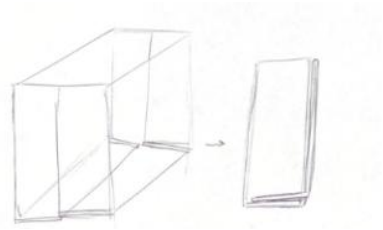
Concept 1 (Figure 25) was designed with the thought that the compartment could fold flat against the wheelchair. An issue that arose concerned the hinges; they would need to be mounted on the stand of the wheelchair where the electronics and battery are placed. This could be dangerous, especially if the hinge pulls and breaks near the important components for the wheelchair's operation. Moreover, the box would not be able to fold fully flat because the angle of the backrest would obstruct that position, so a way to keep that box folded in place would need to be provided.

Figure 25: *Mechanism concept 1*



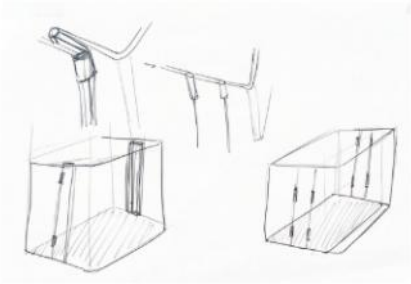
Concept 2 (Figure 26) consisted of having a compartment made of two parts that utilised a rail system to move the parts closer or further apart. Although this seemed like a promising solution, thin rails could not be sturdy enough to support the weight of a cabin bag, so thicker rails would be needed. Additionally, the rails would also need to be collapsible, otherwise they would protrude out of the wheelchair's base when the box is folded.

Figure 26: *Mechanism concept 2*



The next concept (Figure 27) considered a fixed side of the box and a moving one where small hinges would fold part of the compartment inward. This solution would only minimally reduce the box's size. Furthermore, the base of the box would need to either be foldable or be smaller than the opened box size to not protrude under it once it has been folded. Having a smaller compartment base would mean an open slit where the wheels of the luggage could fall.

Figure 27: *Mechanism concept 3*



The fourth idea (Figure 28) followed the shape of the wheelchair's stand that holds up the seat. The benefits of this concept were that the luggage compartment would be able to completely fold into the wheelchair, leaving enough space for nesting. However, the corners closest to the stand would need to be greatly rounded in order to not damage the seat of the wheelchair when unfolding into the open position. In addition, the back of the compartment would be open, and the luggage would roll out. Concept 5 (Figure 29) added an elastic strap to prevent the suitcases from falling off the wheelchair. That strap would be adjustable for larger suitcases and long enough to let the compartment fold back. However, due to the short height of the wheelchair stand these two concepts could not be used, as the suitcase would be at risk of falling over the short edge of the compartment.

Figure 28: *Mechanism concept 4*

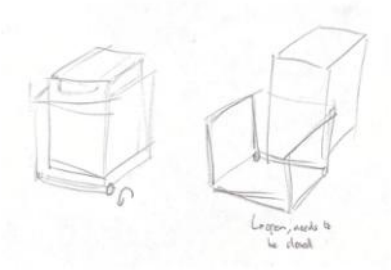
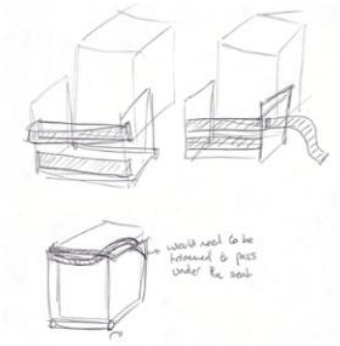


Figure 29: *Mechanism concept 5*



Concept 6 (Figure 30) was inspired from foldable toolboxes (Figure 31). The mechanism was simple, but the way it was set meant that the luggage compartment would double in size when folded, which could not work due to the backrest's inclination. Therefore, Concept 7 (Figure 32) was designed so the compartment would fold horizontally instead. This meant that the box could reduce its size practically in half and allow another wheelchair to move closer for nesting purposes.

Figure 30: *Mechanism concept 6*

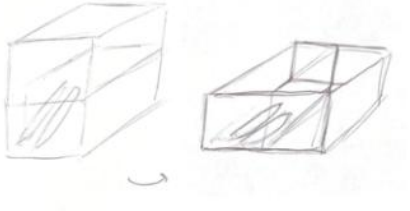
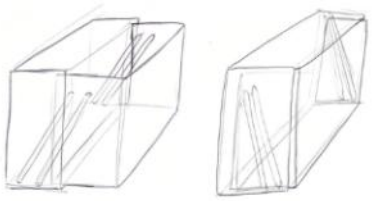


Figure 31: *Foldable toolbox example*



Figure 32: *Mechanism concept 7*



Moreover, Concept 7 requires little force, only one hand and one movement to open and close the luggage compartment. This makes it more accessible as users with limited physical strength or carrying a mobility aid can still easily use it. This complies with the UD principle six of low physical effort, which aims to produce design that can be used efficiently and comfortably with a minimum of fatigue (CEUD, n.d.).

Choosing this last mechanism meant that irregular shapes for the luggage compartment's design could not be used. Therefore, a clean and symmetric shape was chosen. The final design of the luggage compartment, seen in Figure 33, had rounded corners and a curved front to fit with the design language of the seat. The side walls had to remain straight so as to be able to fit both parts of the box together.

Figure 33: *Final luggage compartment design*

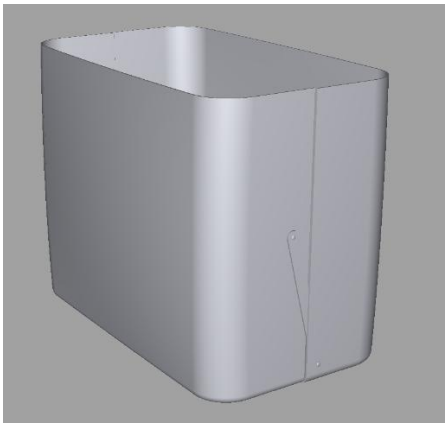
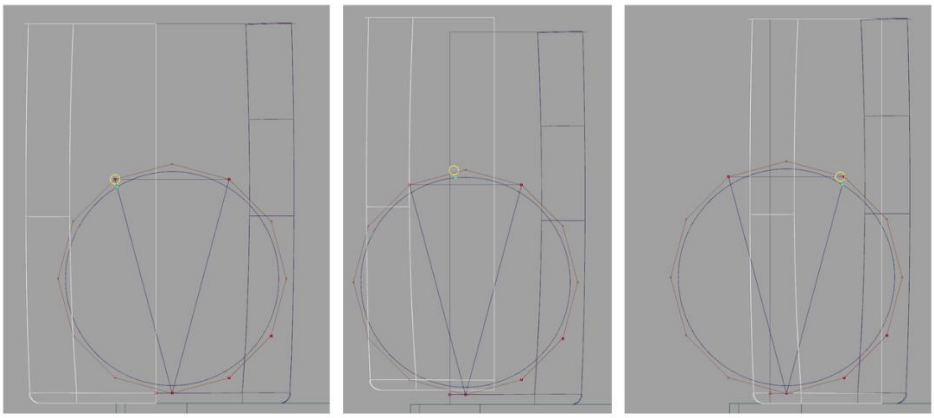


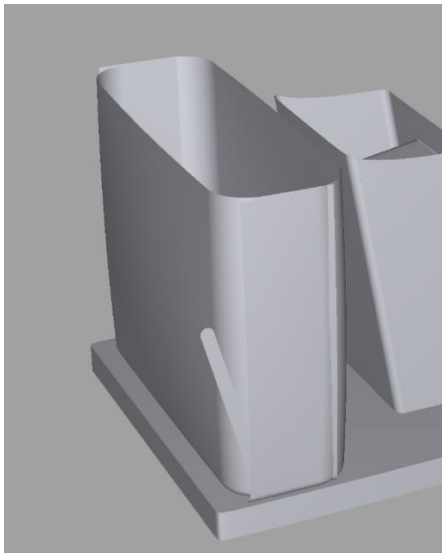
Figure 34 shows the first iteration of the mechanism designed in concept 7. The suitcase compartment was divided into two parts, with the back part (shown on the left side in this figure) being slightly narrower that the front part so as to fit together when folded. The front and back parts had a metal bar attached at a certain point, and since the front part of the box remained fixed, the back part would follow an arc trajectory created by the metal bar to fold inside of the other part.

Figure 34: *Visualisation of folding mechanism for box concept 7*



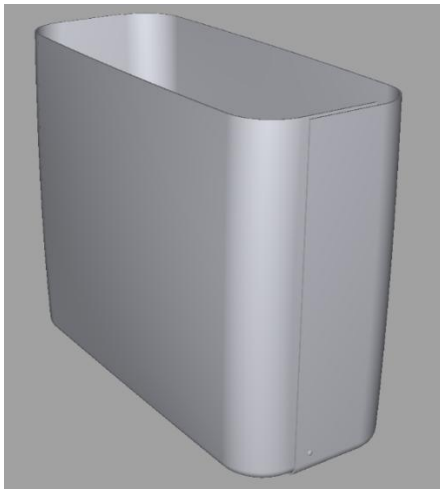
The back part should, however, not end up at the same base level as the front part because each sections have a thickness, so they are not able to physically align. This means that the back compartment piece should lay on top of the front piece when the compartment is closed. Furthermore, selecting where the metal bar should be attached was challenging, because not placing the points correctly resulted in the two pieces overlapping (Figure 35)

Figure 35: *Folding mechanism issue*



After making the changes mentioned above, the final luggage compartment was finished. Figure 36 displays it closed.

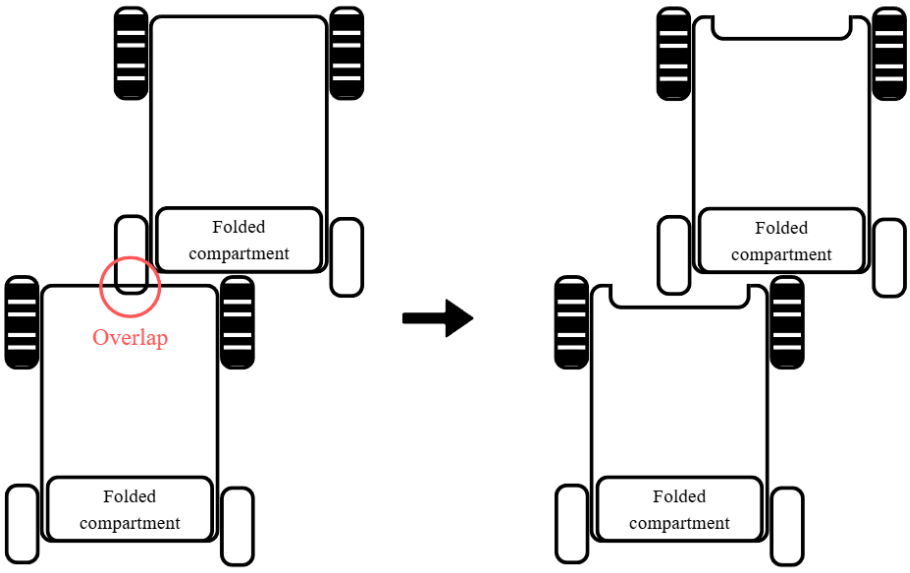
Figure 36: *Final closed luggage compartment*



v. Footrest mechanism

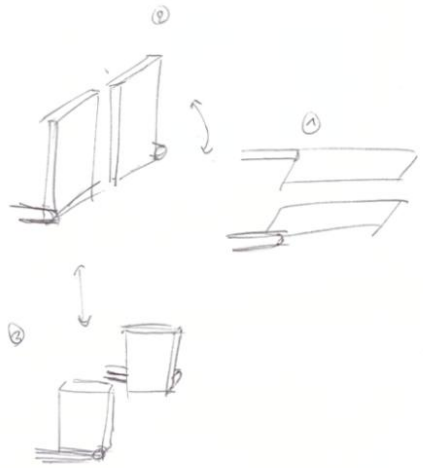
The next step after designing a suitcase compartment that could fold was to focus on the footrest mechanism. As seen on Figure 37, the back wheels overlapped over the base of the wheelchair, but creating a small cutout for them in the base was not enough to solve this issue; a footrest had to be placed where the cutout is, which, if it were made of one solid block, could not allow nesting into the base.

Figure 37: *Visualisation of solution for overlap of wheels and wheelchair base*



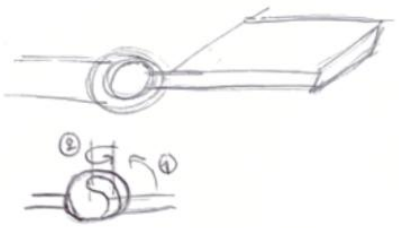
To liberate the space for nesting, the footrest needed to fold out of the way. In order to keep ergonomic footrest dimensions, as mentioned below, it needed to be designed in two parts that could independently fold in two directions: upwards and inwards (Figure 38).

Figure 38: *Footrest mechanism movement*



This motion was achieved with a ball joint, as shown in Figure 39, which allowed each part to move first upwards and then rotate inwards, leaving the cutout of the base unobstructed. Small pins inside the ball joint allowed it to stop at the specified positions.

Figure 39: *Ball joint mechanism*



The size of the footrest was derived from anthropometric data. Jurca et al. (2019) state that the most frequently occurring foot length for European males was 270 mm and 245 mm for females. Usually, footrests in wheelchairs only support the heel and midfoot areas, not the entire foot, which enables the creation of more compact designs. The authors also mention that the mean foot width varies with foot length; for European populations, it is typically around 95-105 mm for males and 85-95 mm for females within common adult foot length ranges. Therefore, the chosen footrest sizes, namely 187 mm in length and 163 mm in width for each foot, provided adequate support for a broad spectrum of users including those in heavy footwear or foot braces. The total width of both footrest parts was 363 mm, which also considered a 37 mm distance between the panels, which allowed for comfortable feet positioning.

vi. Seat design

Following the steps taken in Wheelchair ideation, several criteria from Table 2: *Design requirements set from primary user needs* were considered to design the wheelchair's seat. Large, rounded seats that obstructed transfers and access from the side were not chosen. Moreover, generally bulky designs were not considered as they would not be able to allow compact wheelchair storage. Two separate cushions, for the backrest and seat, were selected for padding and comfort, as well as to be easily removed and changed when needed. The design direction included slow curves, a modern appearance and a connection to waves, as seen in the evolution of sketches in Figure 16 above, it was shown that a seat designed out of one continuous line was explored.

The chosen seat design, illustrated in Figure 40, included slow curves, reminding the user of the shape of a wave, and connected the seat, backrest and handle in one continuous line. It gave the wheelchair a sleek and modern appearance, while staying friendly and approachable.

Figure 40: *Seat design illustration*



The seat was designed to support a posture that balances comfort and stability. The bottom seat was given an upward inclination of approximately 3 degrees to reduce forward sliding and improve pelvic stability. Clinical wheelchair seating guidance suggests that positive seat angles between around 3 to 7 degrees may improve seating stability and comfort during prolonged sitting (KARMA Medical, 2025).

The backrest was inclined 100 degrees from the horizontal axis because a slightly reclined position is considered more effective for reducing pressure on the lumbar spine than a 90-degree position. Barth et al. mention that backrest inclinations between 100 and 110 degrees provide more comfort for wheelchair users and lower spinal disc pressure during extended sitting periods (Barth et al., 2020).

The cushions' colour was carefully thought through. A dark blue colour was chosen to add a more vibrant detail in airport environments while still maintaining a professional and calming appearance. Airlines often choose different shades of blue for their seats because blue is associated with calmness, trust, and reliability, which can help reduce stress (Szweda, 2023). Research in colour psychology has also shown that blue is commonly associated with feelings of peace and serenity, making it suitable for public spaces where users may feel nervous or overstimulated (Cherry, 2026). Additionally, the choice of a dark blue fabric is practical for high-traffic environments. It conceals dirt, wear, and scratches more effectively, which helps to maintain a cleaner appearance for a longer time and lengthens the product's life cycle. This is especially useful in airports, where wheelchairs may be used continuously throughout the day by many different passengers.

The wheelchair also included anti-tip wheels at the back of the base to increase safety for users.

vii. LiDAR pole

DAAV's technical team requested a tall LiDAR pole to be designed and added to the autonomous wheelchair. As mentioned in the Competitor analysis, having a 360 sensor is essential for autonomous navigation. Two planar sensors were also placed below the base to allow for better obstacle detection without the passenger's legs or clothes blocking their view.

Placing the 3D LiDAR higher allows for a less obstructed field of view to map the airport, as it can see above most people's heads. It also enables the wheelchair to be seen from further away, making it safer for passersby.

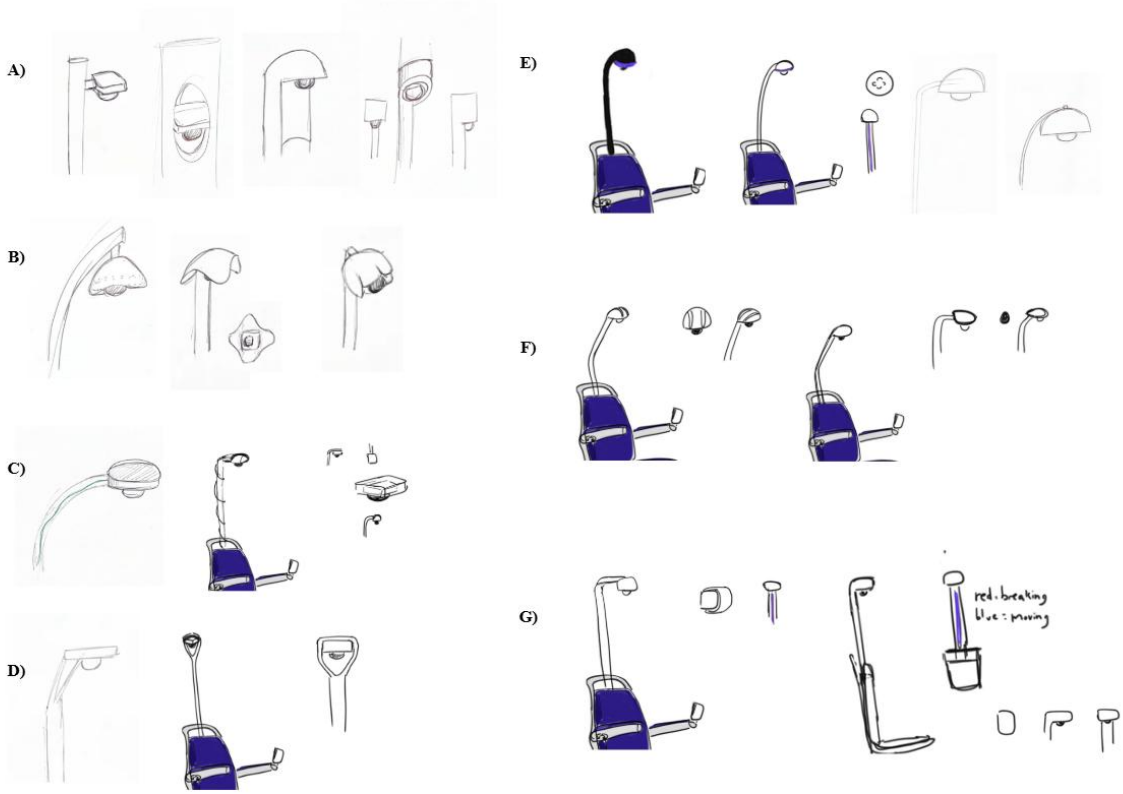
A sketching session was organised, where multiple pole shapes and top shapes were sketched. The poles were drawn first without including the rest of the wheelchair and later also drawn with part of the wheelchair. This allowed to sketch freely and let new ideas develop without restricting them first. The drawings that had similar characteristics were then grouped into categories to be analysed (Figure 42).

The sketches considered the size of a common 3D LiDAR sensor, shown in Figure 41, and its cable that would be used to collect data. They also placed the pole from the backrest, since a previous wheelchair model, seen in Figure 7, showed that a pole on the sides compromised its stability and led to issues with the data collected from the sensor.

Figure 41: *3D LiDAR sensor*



Figure 42: *LiDAR pole sketches categories*



Category A included designs that were too bulky and would obstruct the LiDAR's field of view considerably, so they were not considered for the final stage. Category B was more ornamental and would not align with modern airport environments. The next category experimented with the sensor's wire. A translucent pole with a colourful cable as well as a spiral shaped cable were illustrated, but these solutions were considered too industrial or could compromise the passenger's safety and were therefore not pursued. Group D showed a pole that would hold the LiDAR from of its lateral sides, minimising the obstructed view for the sensor. This approach had a modern look but could be perceived as overly technical and less approachable.

Category E was inspired by floor lamps and followed a rounder design language, with one of the sketches being inspired by a moon jellyfish to recall the theme of the ocean. Although these concepts were quite friendly and pleasant, their look was too similar to lamp shades, so they were not considered. Category F was more daring. The designs for the top holder were bold; they included stripes and a tear shaped holder with bent poles. Despite being innovative, their appearance was deemed less approachable and could also be interpreted as sporty.

The last category, category G, pursued a mix of squared and rounded shapes. Previously, on the company's current wheelchair models, the LiDAR pole was extended and placed right above the seat, because this would maximise the sensor's field of vision around the wheelchair. However, DAAV's team agreed that moving the LiDAR slightly backward would not compromise much the field of vision. Therefore, it was moved closer to the bar, which made it look more integrated and stable. The last variation on the right was chosen for implementation because of its modern look and rounded features that fit with the wheelchair's design language. A light strip was also added at the back of the pole to improve the wheelchair's visibility and increase safety for bystanders, as well as to give the wheelchair a futuristic edge.

viii. Mobility aid holder

As mentioned in Try it yourself (4.2.2) and the Semi-structured interviews (4.2.4), a mobility aid holder closer to the seat would benefit the accessibility of users with reduced mobility when sitting down and increase the inclusivity of the wheelchair. Thus, a holder integrated into the armrests could be an efficient place for that purpose.

So as to safely retain long canes and crutches, the holder needed to support both the end and the middle of the mobility aids. To minimise material and improve the appearance of the holder, a cup near the bottom base of the wheelchair and a holder near the armrest would provide sufficient stability for the assistive equipment.

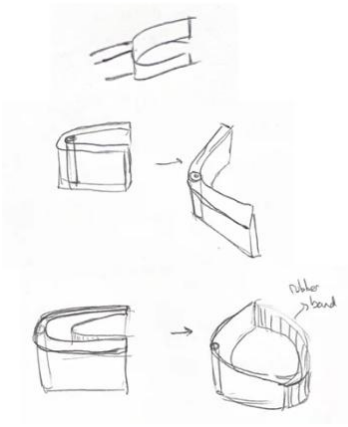
Choosing a higher cup, as seen in Figure 43, would increase material use. A shorter cup with a rim to prevent the mobility aids from slipping out would be sufficient. Moreover, having a completely round design meant that an extension would need to be made to connect the cup to the base, so a flat back was designed to allow the cup to be mounted closer to the wheelchair's base. The size of the cup was chosen so as to fit two standard crutch tips of around 16mm in diameter.

Figure 43: *Cups for mobility holder*



Concealing the mobility holder when needed could improve the wheelchair's appearance, so the top part of the holder would need to be integrated into the armrest. To do so, a small portion of the armrest would be hollow and could open with a small door to reveal the mobility holder. A few clamp solutions were considered in Figure 44. A clamp design that would be able to expand size and hold two crutches instead of only one cane was experimented with. The third solution on the bottom was chosen, as it could fold its rubber part to reduce its size and fit into the armrest compartment, as well as expand outwards when needed to fit multiple assistive devices.

Figure 44: *Clamps for mobility holder*



ix. Electronic device

To share the user's destination with the autonomous wheelchair and use the software provided by the company, an electronic device needed to be added to the product.

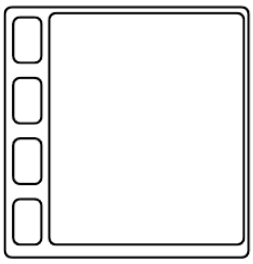
This device required to have a screen to display information for the user. The screen dimensions of around 79 mm by 69 mm were chosen to have a compact device but keep the legibility clear. Having a simple, highly legible interface with the possibility of changing the text size would be beneficial for the users. The device also included a back camera for future software implementations, such as boarding pass scanning.

To reduce the interaction complexity, large physical buttons were preferred over a fully touchscreen interaction because tactile controls can be located and operated through touch alone, reducing dependence on visual attention. This complies with the universal design (UD) principle 4 for perceptible information, which aims to convey information regardless of the user's sensory abilities (CEUD, n.d.). This reduced the reliance on precise screen interaction and was beneficial for users with visual impairments or users that have reduced fine motor skills, thus limiting the errors the users could make by only using the touchscreen function alone (principle 5 of UD: tolerance for error) (CEUD, n.d.). It also reduced cognitive load by limiting the elements that needed to be interacted with.

Several concepts following these criteria were designed.

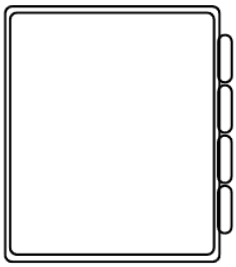
The first concept (Figure 45) considered four squared buttons on the side of the screen. This made the device wider and positioning the buttons on the left-hand side of the screen would mean that, because the device is placed on the wheelchair's left armrest, people using their right hand would need to reach further to press the buttons. This complies with the seventh principle of UD for size and space for approach and use, which suggest placing components for easy reach (CEUD, n.d.).

Figure 45: *Device concept 1*



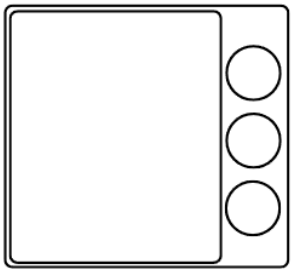
Concept 2 (Figure 46) explored the possibility of placing the buttons on the side of the electronic device. Although this created a modern appearance, it made it more difficult for the buttons to be seen. This would be especially hindering if the users needed to see a graphic on the face of the buttons, as plain ones would create more cognitive load on the users when trying to guess each button's function.

Figure 46: *Device concept 2*



Concept 3 (Figure 47) placed the buttons on the right-hand side. The first concept had four buttons, but having more buttons reduced their overall size and created more choice options. To operate a simple interface three buttons are deemed enough. Indeed, these buttons allow the user to move inside the software and choose amongst functionalities there, but for the physical buttons, the choices: left, right, and accept, or up, down, and accept can be easily used and understood.

Figure 47: *Device concept 3*



To reduce the width of the electronic device and to permit users to access the buttons in an easy and visible way, the next two concepts placed the buttons right under the screen.

Concept 4 (Figure 48) used large, squared buttons, while concept 5 (Figure 49) used large, round buttons.

Figure 48: *Device concept 4*

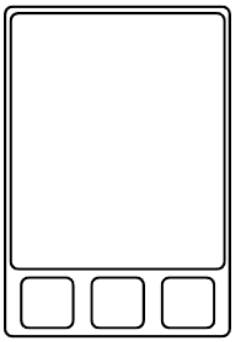
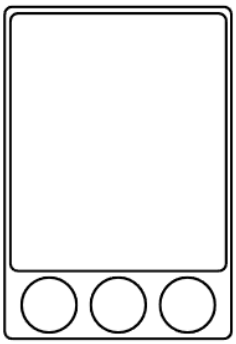
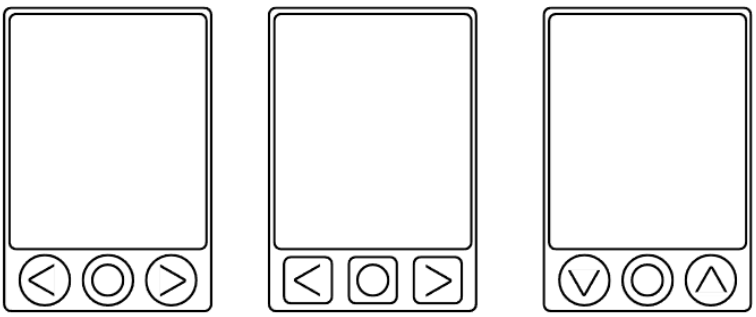


Figure 49: *Device concept 5*



Before selecting from the last two concepts, haptic details needed to be chosen and added to the buttons' surface. Adding a three-dimensional graphic provided tactile feedback for people with visual impairments and further clarifications for all users. As mentioned above, the graphics should convey options to accept, move up, down, left or right. Multiple phone companies and game controllers use a circle to convey a confirm or accept command. Therefore, to reduce cognitive load and make the graphics intuitive and easy to understand, the circle was chosen for that effect, following the third principle of UD of simple and intuitive use (CEUD, n.d.). Figure 50 shows the different graphics placed on the buttons.

Figure 50: *Options for the haptic button details*



Having the arrows pointing up and down was less visually attractive and using the left and right arrows could still allow to effectively move through a software interface. The

circle in the middle also created a more cohesive design when used with circular buttons, so the first concept on the left was chosen. This device was made with a pleasing and cohesive design, while also complying with universal design principles and improving inclusivity and accessibility.

4.4 Deliver

4.4.1 Final design proposal

The autonomous wheelchair was 3D modelled in the software Autodesk Alias. It was created using the designs and constraints mentioned in the previous chapters. Figure 51 shows the final design proposal in its entirety.

Figure 51: *Final design*



Below, in Figure 52, some moving parts are shown, and their positioning are explained by using photoshop blur and opacity effects. Both armrests can lift entirely and let the passenger sit from the side if they wish to do so. The footrest folds along two axes, which also liberates space in the front of the wheelchair to sit more comfortably. The footrest and back compartment positions when folded can be seen in the middle image.

Figure 52: *Final design: moving details*



When looking at the back of the wheelchair, the light detail on the pole can be seen (Figure 53). It could turn red when braking or driving, as well as white, which can be a calmer colour. The wheelchair appears modern and the light detail adds a futuristic touch.

Figure 53: *Final design: back light*



Figure 54 showcases the mobility holder with the example of a cane. If observed carefully, the detail on the back wheel cover can also be seen. This detail resembles a drop of water and is a reminder of the wave inspiration and red thread of this project.

Figure 54: *Final design: wheel cover and mobility aid holder*



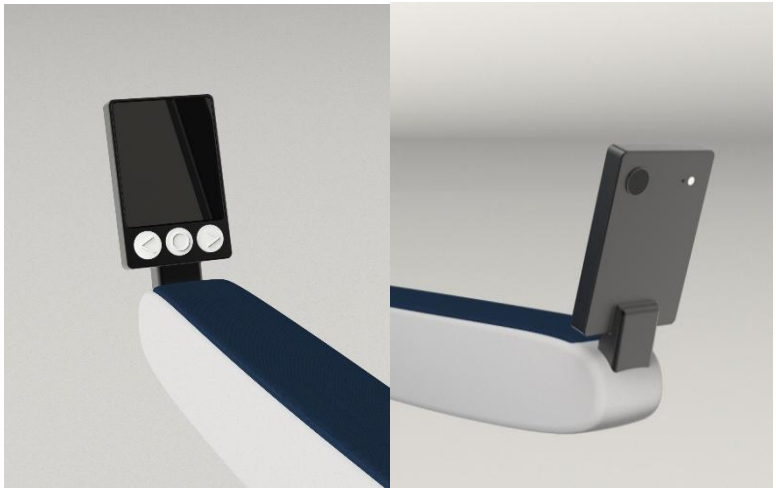
Next, is a demonstration of the partial nesting mechanism explained in 4.3.4iii Wheelchair storage solutions (Figure 55).

Figure 55: *Final design: partial nesting*



Figure 56 shows the buttons and screen of the electronic device, as well as the back of it, where a camera and flash can be used if needed.

Figure 56: *Final design: electronic device*



Lastly, a contextual render of the wheelchair in a modern airport was generated using artificial intelligence and photoshop to perfect the details (Figure 57).

Figure 57: *Contextual render*



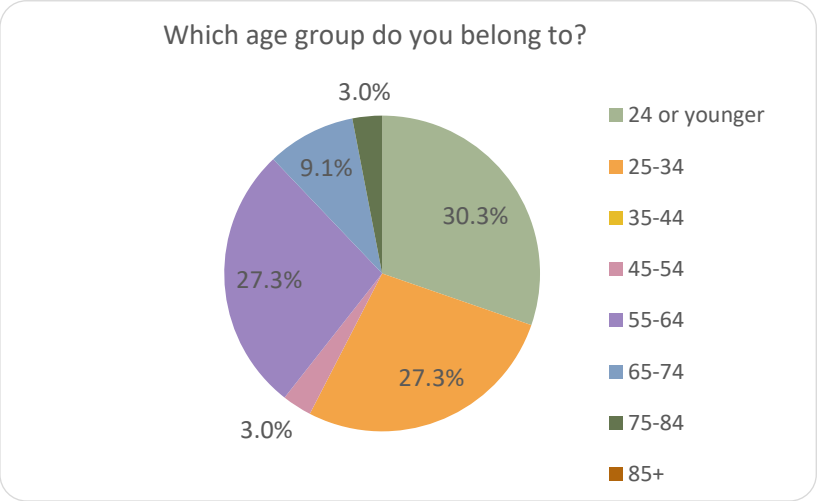
4.4.2 User validation survey

To further validate the design proposal and get additional insights, a user survey was conducted. The survey followed the ethical guidelines used for the semi-structured interviews and therefore allowed the answers to be used in this study. The survey contained 17 questions and collected the users' previous experience with mobility assistance and their perceptions of the final design proposal, as well as an open section to suggest improvements. The survey was sent to participants through online groups and encouraged people with reduced mobility as well as other users to reply, which led to a total of 33 participants to answer the survey.

The survey was divided into three themes: participant context, overall design perception, and accessibility and interaction. The questions utilised a 6-point Likert scale, which removes the neutral midpoint and ensures the respondent's cognitive engagement, which can lead to more meaningful opinions (Krosnick et al., 2002).

The first three questions (Figure 58, Figure 59, Figure 60) focused on the participant context. Results showed a varied age group, with the majority of respondents aged 34 and lower, closely followed by the age group of 55-65, then 65-74. This is important to keep in mind for the analysis of the results, as the younger age group may usually only be a target user in case of a possible injury, therefore they may not be familiar with mobility issues or wheelchairs and may answer the questions by making certain assumptions depending on what they imagine a PRM user to be like. However, having such a varied age group can be positive as they may bring different perspectives and perceptions to the results.

Figure 58: *Survey question 1*



In Figure 59 and Figure 60, results showed that around 80% of participants have never used mobility assistance, which once more suggests that the questions that concerned people with reduced mobility specifically might be answered with assumptions such as what a PRM might be able to do or feel. On the other hand, around 20% did use the service in the past, which adds a non-negligible number of opinions from real PRM users. Furthermore, the age groups having positively answered this question were part of the 24 or younger, 25-34, 55-64, 45-54, and 65-74 age groups. This great variety

implies that the results from PRM users themselves are not age dependent, as they all belong to a large age spectrum.

Figure 59: *Survey question 2*

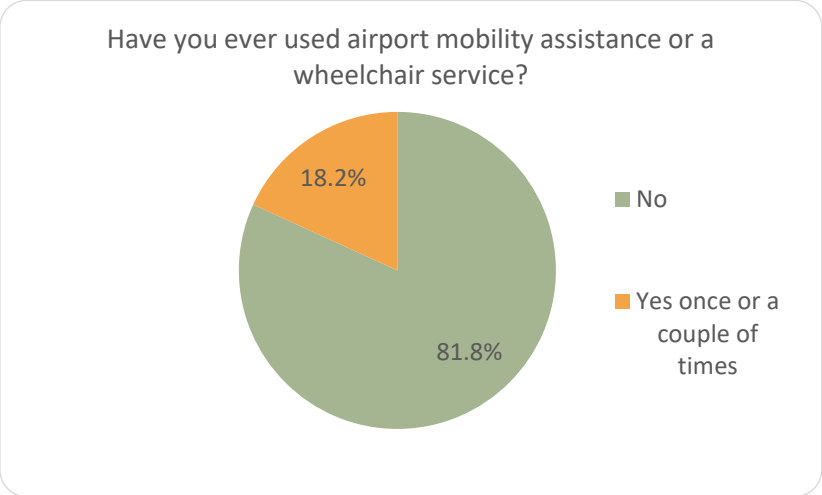
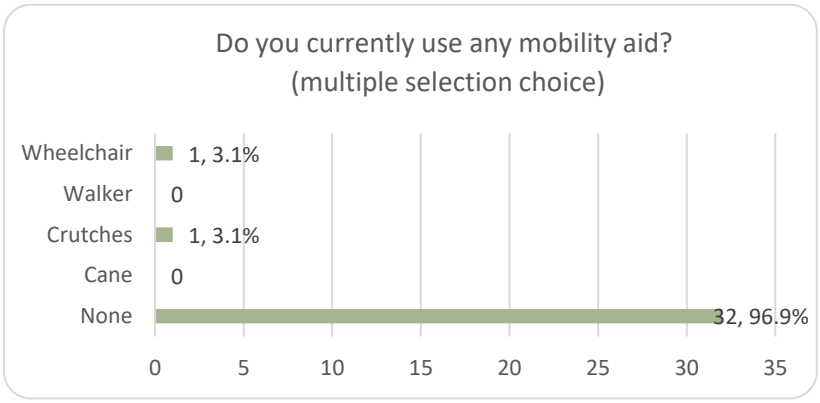


Figure 60 shows a low percentage of mobility aids checked. This can mean that the participants having used mobility services in the past might have been only temporarily injured or had a reduced mobility that did not necessitate the use of mobility aids. Another possibility is that they do not need mobility aids anymore but did use them at the time they used the service, but since the questions mentions a current use, they answered the questions negatively. These results therefore might not be an accurate representation of participant's use of mobility aids but do signal that a small percentage was able to answer the question related to mobility aids from a personal experience.

Figure 60: *Survey question 3*



The rest of the survey used a 6-point Likert scale, where 1 corresponds to "strongly disagree" and 6 to "strongly agree". Questions four to seven focused on the overall design perception of the autonomous wheelchair and presented the images in Figure 61, with the following text when starting: "Below are images of the autonomous wheelchair that includes a pole with a sensor for navigation at the top, a luggage compartment at

the back, a screen with buttons, and an integrated mobility aid holder in the armrest. The last picture explains that both armrests and the footrest can fold completely up."

Figure 61: *Survey image of the full wheelchair*



As Figure 62 shows, over 72% of participants felt strongly that the wheelchair's design was visually approachable. Moreover, Figure 63 shows an overwhelming number of participants strongly agreeing with the statement. This validates the intended design vision which aimed to appear friendly and approachable, as well as have a modern design that would integrate well in airports.

Figure 62: *Survey question 4*

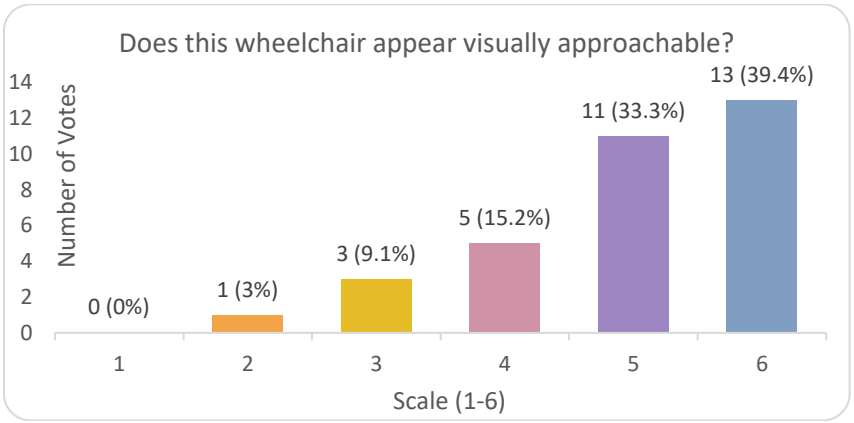
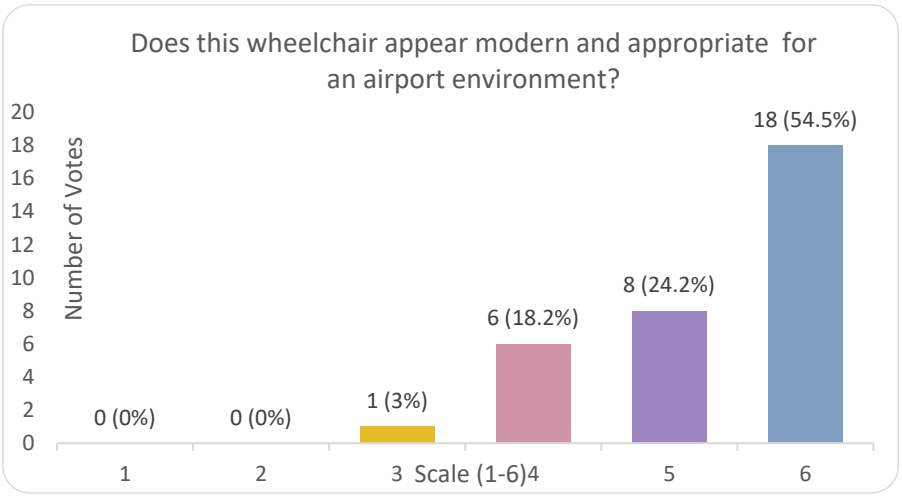
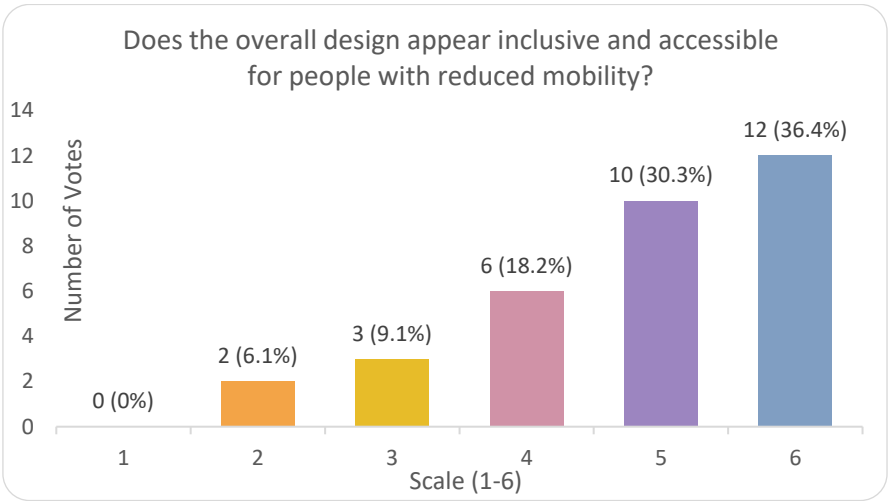


Figure 63: *Survey question 5*



Considering the appearance of the wheelchair with regards to inclusive design for PRM users, around 84% of respondents were of a positive opinion (Figure 64). This can suggest that the design did not seem overly complicated to use or did not have any visible features that would exclude people with reduced mobility, and that it indeed appeared accessible.

Figure 64: *Survey question 6*

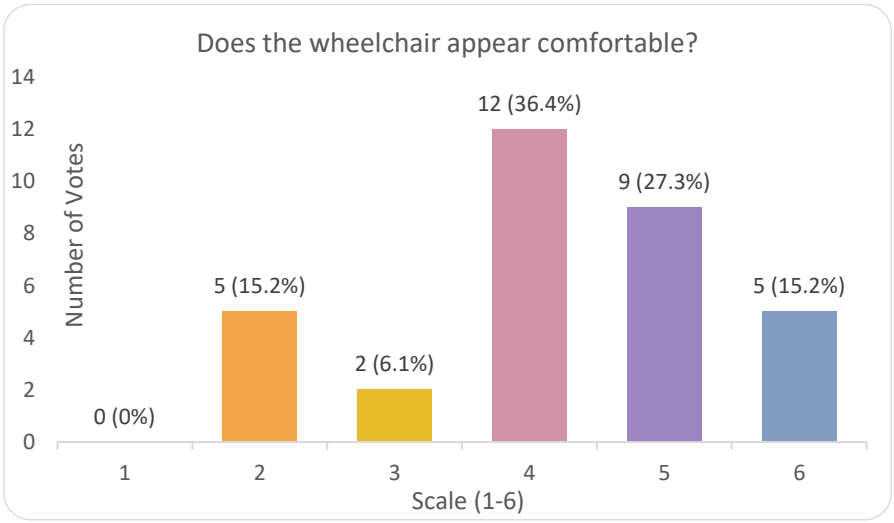


Question 7 (Figure 65) generated some mixed responses. Although the majority of participants perceived the wheelchair as comfortable, around 36% slightly agreed and five participants disagreed. The thickness for the wheelchair's cushions was around 97mm for the seat cushion and around 87mm for the back cushion, which is plenty to support users for their journey in the airport (Ragan et al., 2002). Therefore, the seat design, which curves upward hiding the cushions from the side, might have mislead participants into believing the cushions were thinner and less comfortable.

The five respondents that disagreed were all from the two youngest age groups and had never used a wheelchair service, which may mean that they are not familiar with the thin leather or material strap seats often used in those services. If participants are not familiar with regular wheelchairs it is also easier to compare the proposed design to

traditional office chairs or thickly padded house chairs, instead of a traditional wheelchair.

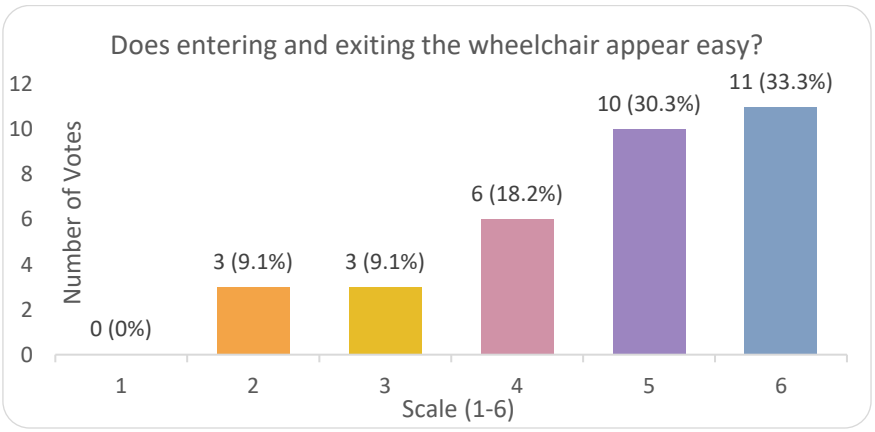
Figure 65: *Survey question 7*



The remaining graphs concern questions about the wheelchair's perceived accessibility and interaction.

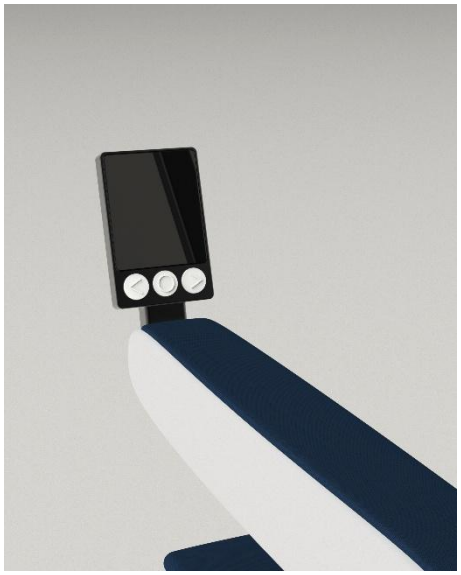
Question 8 (Figure 66) asked about the seating access design. Three respondents disagreed and a couple of them added a comment in the open feedback section mentioning that an armrest should be able to lift to allow access from the side, which is shown in the picture already. Therefore, it is possible that some participants skipped the explanative text or looked at the images too fast. However, around 63% of respondents strongly agreed that it seems easy to enter and exit the wheelchair and 18% more also answered positively, which shows that providing side access with movable armrests had a positive outcome for accessibility.

Figure 66: *Survey question 8*



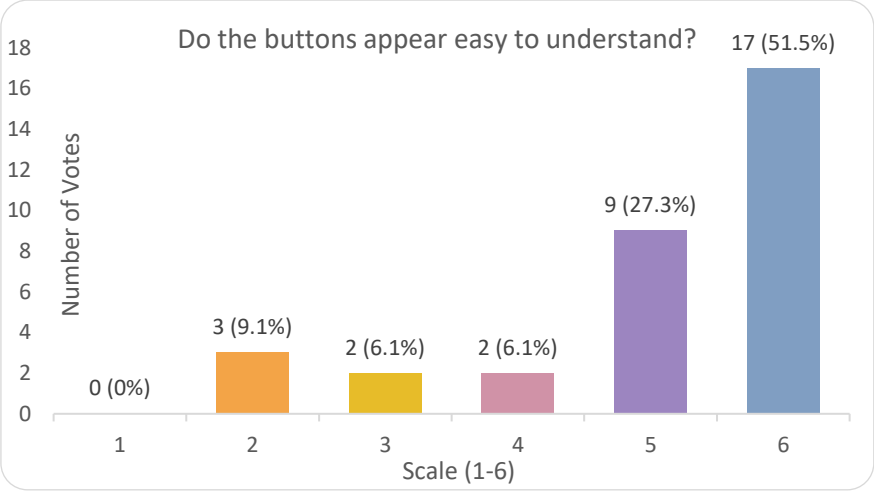
For the next questions, a closer image of the electronic device and its buttons was shown in the survey (Figure 67), with the following text: "Here is a closeup of one of the wheelchair's features, which includes buttons with 3D raised graphics."

Figure 67: *Survey image of the buttons closeup*



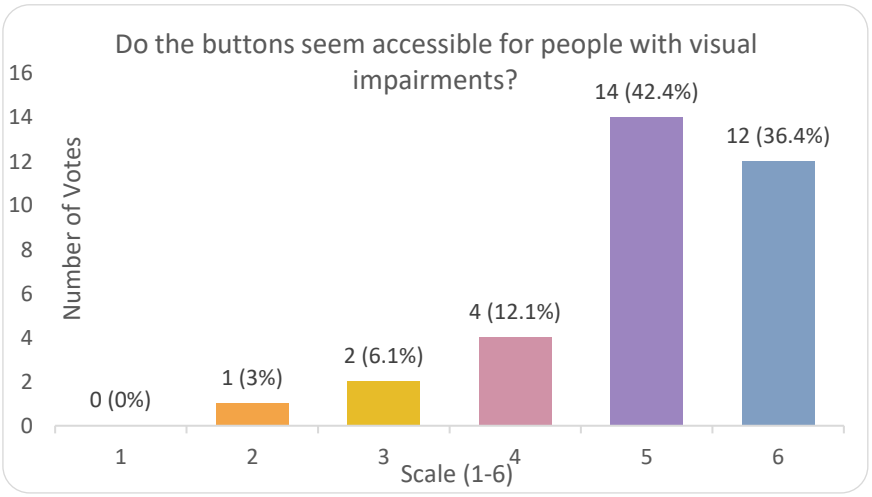
An overwhelming majority of participants strongly agreed that the buttons appear easy to understand (Figure 68), which strongly validates the design proposal of using simple symbols to communicate the functions of the buttons to reduce cognitive strain.

Figure 68: *Survey question 9*



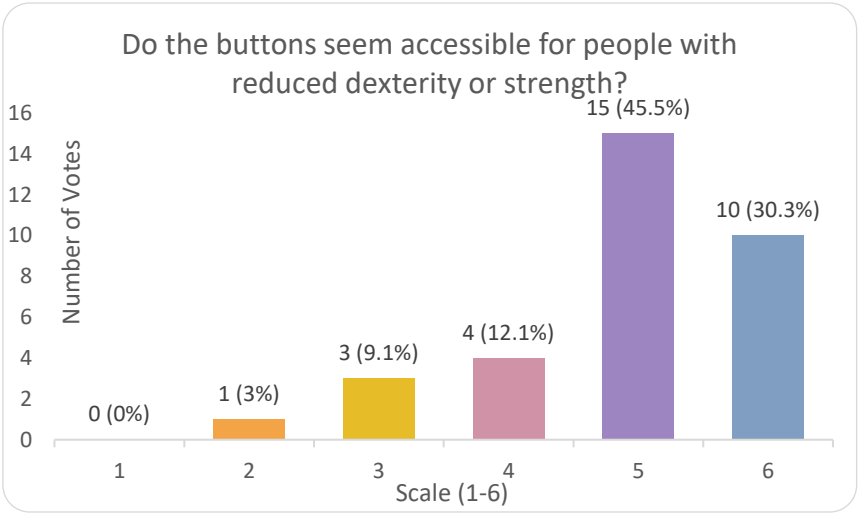
Question 10 (Figure 69) and 11 (Figure 70) asked about the buttons' accessibility. Around 78% of respondents agreed that the buttons seem accessible for people with visual impairments. Indeed, they were designed with a tactile graphic for that intent. A comment from a user suggested that making the graphics more visible by changing their colour would help see them better for people with visual impairments. Although the initial proposal suggested moulding them in one single colour to reduce cost and produce less material waste, adding a layer of colour in post-processing would improve the visual feedback of the buttons. The size of the buttons was also chosen with the intention of making them easier to see and push.

Figure 69: *Survey question 10*



Furthermore, around 45% of participants seemed to agree with question 11, that the buttons appeared accessible for people with reduced finger dexterity. Indeed, their size was not only chosen for people with visual impairments, but also to facilitate pressing the button chosen and not one on the side, for example, if they were designed smaller. A participant that disagreed, suggested to keep the buttons and to add a joystick, as users with sever hand or finger dexterity could still not be able to press buttons.

Figure 70: *Survey question 11*



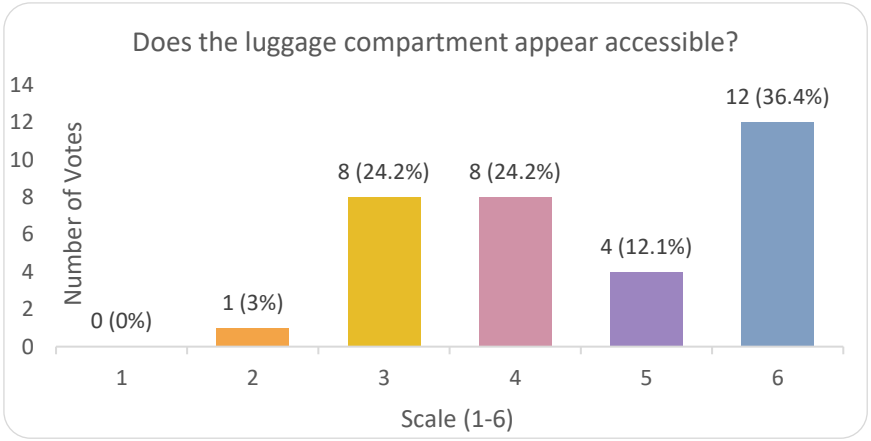
Question 12 and 13, shown in Figure 72 and Figure 73, are similar questions about the luggage compartment. The following image was presented in the survey (Figure 71).

Figure 71: *Survey image of the mobility aid holder and luggage compartment*



The first one was more general and around 36% of participants strongly agreed that the compartment appears accessible. However, around 48% believe it to be less accessible. Multiple participants later requested the baggage compartment to be made bigger or more flexible as they believed that the shape was too restrictive. The luggage compartment was designed slightly larger than airline's hand baggage restrictions in mind, however, this still means that the luggage should not be larger when boarding the aircraft. Nevertheless, many users tend to overpack and fill their luggage more as some airlines do not conduct strict size verifications. In this case, this frustration might stem from the fact that this luggage compartment would expose users carrying a larger bag, so providing them with a more flexible compartment could improve their comfort and ease of mind when using this autonomous wheelchair.

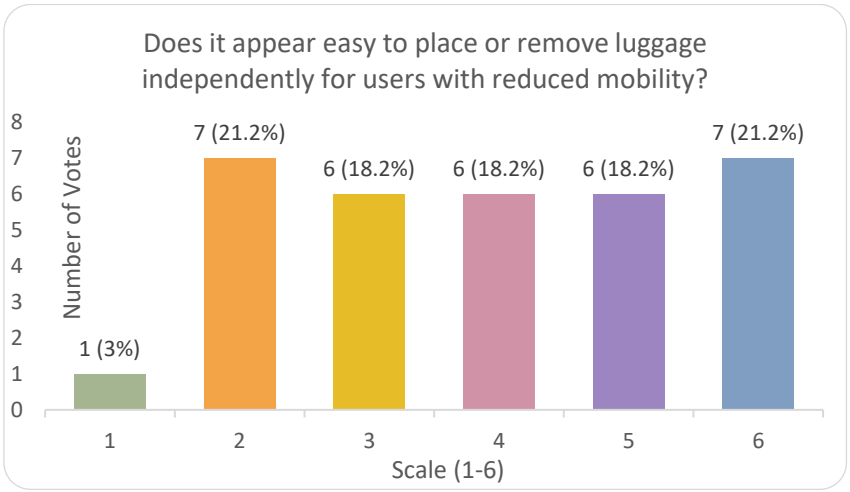
Figure 72: *Survey question 12*



Question 13 collected the most varied responses. The mean value obtained was 3.91, meaning it is above the average score, but that many participants still disagreed with the question. The introduction to the survey mentioned PRM users as: "*mainly elderly passengers, individuals who cannot walk long distances and those who require assistance navigating complex airport environments, which retain enough cognitive and physical abilities to be able to use a service such as an autonomous wheelchair independently.*"

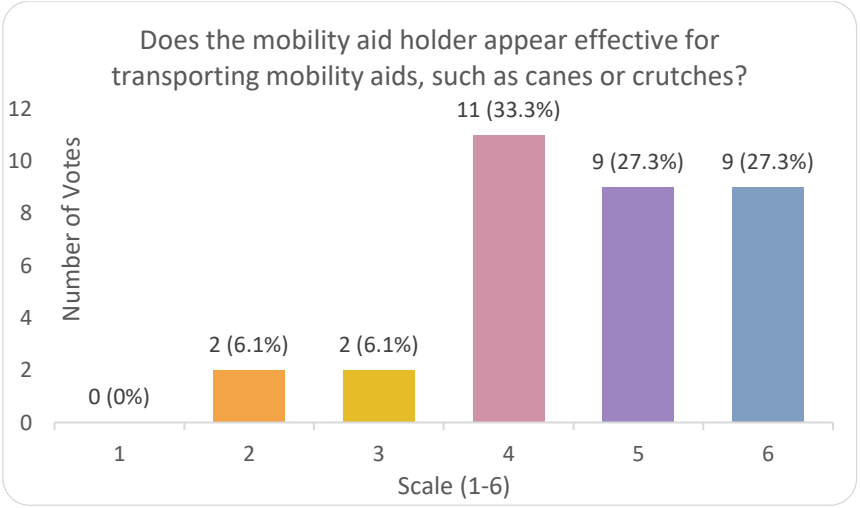
Since this category of users remains quite broad, it is understandable that many participants understood this term differently. PRM could indeed mean a healthy user with one broken knee just as much as it could mean an 85-year-old with arthritis. Since reduced mobility encompasses a spectrum of impairments the distributed responses for this question are natural. The positive responses show that for users that maintain enough mobility to lift objects independently, this luggage compartment can facilitate this movement and proves to be accessible. On the other hand, comments on the open feedback question from the participants that disagreed suggested adding a feature to the compartment which would allow users to place their luggage inside without lifting it themselves. This could be a great improvement to increase the inclusivity of the wheelchair.

Figure 73: *Survey question 13*



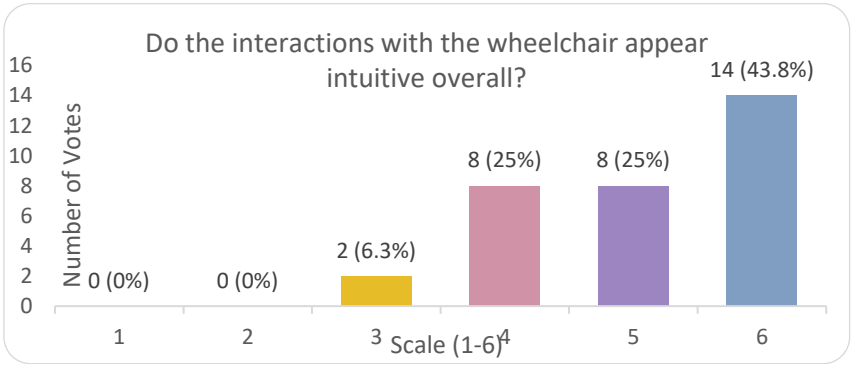
The next question (Figure 74) concerned the mobility aid holder. Although more than 54% of respondents quite strongly agree with the question, around 33% seem to be a bit less sure of its apparent effectiveness, which can suggest room for improvement.

Figure 74: *Survey question 14*



The last question (Figure 75) asked about the overall perception of the wheelchair regarding the intuitiveness of the interactions with it. A strong 43.8% of participants strongly agreed. This is a very positive result and suggests that the goal of reducing cognitive strain by simplifying the functions and interactions with the wheelchair was met.

Figure 75: *Survey question 15*



The last section invited survey participants to give open feedback and suggest improvements for the autonomous wheelchair's design. Apart from the comments mentioned earlier in this chapter, a few additions were recommended. A couple of comments pointed out that a seatbelt would maintain the passenger safe in the unlikely result of a crash or sudden brake. Concerning the comfort, extra padding was requested, as well as a possible addition of a headrest in case a user would prefer to relax a bit more. For the luggage compartment, a way to lock the luggage in the compartment was mentioned, to prevent thieves from helping themselves to the passenger's belonging. These additions and the ones discussed in the previous questions would contribute in a future research to the improvement of the user's comfort, safety, as well as in making the wheelchair more inclusive for people with varying abilities.

4.4.3 Prototyping

The final design proposal was developed into a smaller scale 3D printed prototype. The goal was to create a moving wheelchair by using a race car chassis as a base and attaching the body of the wheelchair on top.

A remote-controlled race car was bought and disassembled to keep the most internal chassis base and suspensions for the wheels were changed to fixed bars to prevent the car from bouncing (Figure 76).

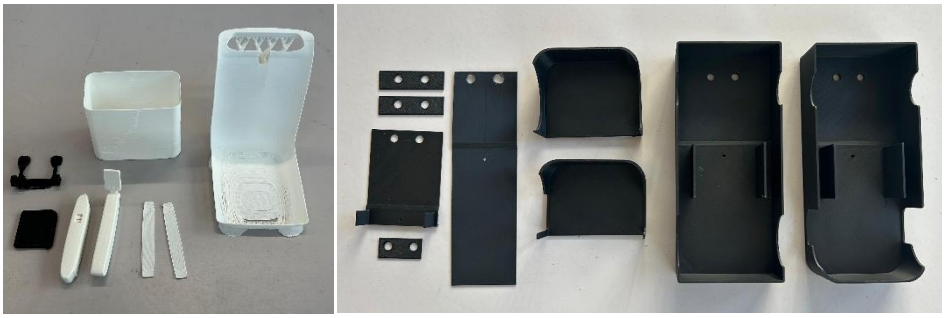
Figure 76: *Race car chassis*





To fit a flat base on top of the car chassis, its dimensions were measured and multiple test parts, adjusting the length and fit of the cover were generated until successful. An opening was also created for the front wheels to rotate. The wheelchair parts were scaled down to 22% and printed on a 3D printer. Several iterations were needed as scaling and positioning on the print plate caused imperfections (Figure 77).

Figure 77: *3D printed wheelchair parts*



After sanding and spray painting the parts to achieve the desired look, they were assembled. Although the car base electronics led to a thicker wheelchair base than the final design concept, the result was a pleasant and interactive prototype to showcase the main wheelchair design (Figure 78).

Figure 78: *Final moving 3D printed prototype*



5 Discussion and Conclusion

This chapter provides a summary of the study's findings and ends with suggestions for further research.

5.1 Discussion of research questions

RQ1: *What are the key requirements for the development of an autonomous wheelchair for people with reduced mobility in airport environments?*

The findings of this thesis reflect that developing an autonomous wheelchair for airport environments requires more than technical capabilities alone. Although autonomous movement remains a crucial functionality of the product, research showed that the success of such wheelchairs was also dependent on how users physically and emotionally experienced the device within airports. The requirements identified in this study therefore extend beyond an engineering viewpoint into a more human-centred perspective on mobility assistance.

A significant finding concerned the importance of independence and autonomy. Throughout the observations and semi-structured interviews, several passengers expressed frustration regarding the dependency on staff created by existing PRM assistance services. In particular, users with temporary injuries preferred to use wheelchairs as a mobility aid without requiring the monitoring of staff. At the same time, some elderly passengers or visually impaired interviewees valued communication and human interaction. This showed that autonomous wheelchairs should not replace airport staff but rather function together in a way that accommodates user preferences and levels of autonomy.

The handling of personal belongings and mobility aids also came up as a recurring issue. Existing airport wheelchairs often required users to carry their belongings on their knees or to balance their luggage between their legs in unstable ways. Therefore, dedicated compartments for both luggage and mobility aids became necessary requirements. Additionally, users expressed the need for features that would improve accessibility and comfort, which led to the integration of ergonomics, haptic features for visually impaired users and facilitated side access entry to the wheelchair.

From stakeholders, several requirements were obtained such as features to handle the wheelchair manually, facilitate repairs and charging, sensor placements and compact storage of the wheelchairs. This demonstrated that the design of such product must balance user-centred needs with the logistical realities of airport operations and the wheelchair’s functionality.

Furthermore, the results indicate that aesthetics and emotional perception play an important role in the acceptance of autonomous wheelchairs. A recurring concern amongst interviewees was that the wheelchair would look overly medical or stigmatising, with most participants preferring a more friendly and modern appearance.

Collectively, the research shows that the key requirements for autonomous wheelchairs in airport environments can be grouped into several categories: independence and autonomy, comfort and ergonomics, accessibility and inclusivity, and storage and

practical usability, which continuously overlap. The findings reinforce the importance of human-centred design within assistive mobility systems and demonstrates that autonomous wheelchairs should be understood as experiential products that shape how users feel, behave, and move within public environments.

RQ2: *How can these requirements be translated into the design of an autonomous wheelchair for PRM users in airport environments?*

The second research question reflected the design process behind the final product and focused on how the identified requirements could be translated into the physical design of an autonomous wheelchair for airport environments.

The requirements were translated into the wheelchair's design through an iterative process, which combined user research, competitor and contextual analysis, as well as inclusive and ergonomic considerations. The final proposal aimed to address practical, emotional, and accessibility related challenges that the PRM experienced when using mobility assistance services in airports.

A clear example of this translation process can be seen in the development of the hand luggage compartment. The study showed that many users struggled supporting their personal belongings while seated in the existing airport wheelchairs. This need was then translated into an integrated luggage compartment by using requirements such as luggage dimensions, accessibility, and functionality and an iterative design process. At the same time, airports required to improve storage efficiency, which led to develop a foldable mechanism which could allow partial nesting to store the autonomous wheelchairs more compactly.

Finally, the project demonstrated that translating user requirements into an assistive mobility product is not a linear process, but rather one involving constant compromise between usability, technical feasibility, operational constraints, and aesthetics. Through iterative solutions and developments, the wheelchair design gradually evolved into a proposal that attempts to support not only efficient transportation, but also inclusivity, comfort, and independence for PRM users.

5.2 Conclusion

This thesis aimed to explore how autonomous wheelchairs for people with reduced mobility in airport environments can be designed through a human-centred perspective. Through observations, semi-structured interviews, competitor and contextual analysis, and iterative design development, the study identified the key needed requirements. Such requirements include comfortable and ergonomic seating, accessible transfers, tactile accessibility, integrated luggage and mobility aid storage, features to support autonomous movement, battery and maintenance accessibility, compact wheelchair storage, and a modern and friendly visual design language.

The project further showed that these requirements can meaningfully impact the physical design of autonomous wheelchairs by shaping features emerging directly from user-centred insights and airport contextual analysis. The resulting wheelchair design

proposal therefore attempts to balance operational efficiency with qualities that support a more inclusive travel experience.

In conclusion, the thesis highlights the importance of autonomous wheelchairs not only as technical mobility devices, but also as products that influence how users feel and experience assistive mobility services. By integrating human-centred design principles into the development process, autonomous wheelchairs may contribute not only to more efficient airport operations, but also to more dignified, comfortable, and independent travel experiences for passengers with reduced mobility.

5.3 Future research

It would be beneficial to pursue the human-centred research by conducting user testing with a 1:1 scale functioning prototype. Real-world testing in airport environments could provide additional insights in ergonomics, accessibility, and usability more accurately than renderings or conceptual evaluations alone. For instance, as mentioned in the user validation survey, the luggage compartment development, whose accessibility had to be compromised with airport requirements and could benefit from additional inclusive features. The mobility aid holder could also be improved by placing the canes from the side of the holder instead of the top.

Shifting the focus to the airports as a client could also provide new interesting perspectives. Since in this case, the main stakeholders would be the airports and not the users, new ways of storing the wheelchairs more compactly could, for example, be researched.

Future research could also investigate how autonomous wheelchairs may adapt to different categories of PRM users with varying physical, cognitive, or sensory needs. The findings of this thesis showed that users do not all seek the same level of autonomy or assistance. Exploring adaptable features or a hybrid assistance system with airport staff could therefore contribute to more inclusive mobility services.

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7 Appendices

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7.1 Appendix A: Passengers semi-structured interviews

Question 1

Have you ever used a mobility assistance service in an airport?

Interviewee	Answer
1	Yes
2	Yes
3	Yes
4	Yes
5	Yes
6	No
7	Yes
8	Yes
9	Yes
10	No
11	Yes
12	Yes
13	Yes

Question 2

What function(s) did you feel were missing from the airport wheelchair?

Interviewee	Answer
1	nothing
2	it varies on the needs, maybe a place to put my cane or my bag.
3	I put my purse on my lap but having a place to put it would be good.
4	My mother is not independent, so I have to carry the luggage all the way to the prm service and bring my mother with her wheelchair here. It would be good to have someone help from the time I arrive to the airport.
5	Cane holder since I'm always carrying my cane with me. I usually have to put my bag on my lap, so a place for the bag.
6	Maybe somewhere for my bag or to hold my water bottle
7	A place for a bag and crutches.
8	(doesn't use a wheelchair, just the prm staff assistance)
9	When the journey is short it's okay to keep my bag on my lap, I guess if it was longer, it would be better to have somewhere to put it. If it's a longer journey it's also important to sit comfortably.
10	I broke my knee so I can't put my bag on my lap now can I? Oh, and something for my crutches.
11	I don't really know, they seem okay
12	I don't feel like I need anything extra, I don't have a backpack, and I only carry my cane, which doesn't bother me. But if there was somewhere to put it, I would use it.
13	Holder for walking sticks and crutches, luggage compartment, for the chairs to be in good condition

Question 3

Was there a moment where you felt uncomfortable using a wheelchair?

Interviewee	Answer
1	The prm service was slow, I had to wait for a long time
2	
3	Not really, I've had good experiences with the PRM service, except having to wait a while sometimes.
4	She [mother] has to be with me at the counter when I check in the luggage, otherwise they don't let me check it in, so I have to transport everything up to the check in and then go wait in the PRM area later.
5	
6	I haven't used a prm service before, so I haven't had any issues
7	The prm service is usually okay, sometimes I need to wait but I haven't had other problems.
8	
9	
10	
11	Not really, the prm's help is good.
12	There was only one time that I had tried to rent an electric wheelchair to use myself, but there were so many people at the airport, and I wasn't used to it, so I was bumping into people and it was chaotic.
13	Yes. I have encountered wheelchairs that were dirty, small and in poor working condition

Question 4

A self-driving electric wheelchair that brings me to my gate at the airport would help me feel more independent and confident. (strongly disagree 1-5strongly agree)

Interviewee	Answer
1	5
2	4
3	3. I prefer to have someone with me, I feel more comfortable with the company of people, that's mostly why I use the service.
4	2. It depends on the level of independence; my mother is completely dependent on me so it wouldn't really change that.
5	
6	Yes it would be interesting for me
7	
8	Not really, I'm blind so I need help with everything, so I prefer to have that link to people
9	If I was all alone maybe I would feel less safe to go alone, but if my son can be with me and the autonomous wheelchair it would be better. Ah it has been tested to not bump into people? Then yes I would say that could be good. My issue is that I don't see very well, that's why my son started to bring me with prm assistance, but if it was sure to bring me to the right place and it's safe then yes it would be quite interesting.
10	I would prefer an autonomous wheelchair, because I just broke my knee, I don't really need extra staff assistance

11	Yes I would say it could help me feel more independent, since I don't need that much assistance, mostly to carry my bags and to not walk very long distances.
12	I'm a bit old school; I prefer to talk to people so having a prm staff is better for me.
13	5

Question 5

I would prefer a wheelchair that appears: A) Friendly and reassuring B) Medical and safe C) Modern and innovative

Interviewee	Answer
1	C
2	A. Most important thing would be that it is accessible
3	B. I guess it's just what I'm used to, this is what all wheelchairs for elderly look like and it seems more adapted to the situation.
4	C. It depends on the age maybe, but I think this might be good for younger people.
5	
6	Yes the design matters. I don't need to feel comforted in a wheelchair, the fact that it's a wheelchair is already comforting. C. I would prefer a modern look because it speaks to the future.
7	
8	
9	I guess the look doesn't matter much to me, as long as it works well.
10	Yes I would say the design matters. C. I would prefer the modern and innovative one because it looks like it would be more advanced technologically and it would be weird if I used a medical looking one, I would feel out of place.
11	
12	I don't mind the design, as long as it takes me where I need to be.
13	A. A friendly and reassuring look

Question6

Any other features to add or other comments?

Interviewee	Answer
1	If it is an autonomous wheelchair, I would want my legs to be protected to make sure that I don't get hurt if it hits anything. The seat should be comfortable. I also want to hear that I'm going to the right place.

2	I would like the wheelchair to give me information as it is driving, since I'm blind, I want it to communicate with me, to know where I'm going and if I'm going to the right place. Language is important and sometimes communication is difficult with the staff depending on the country. It would be good if it also told flight info and to know whether there is time to get to the gate. Since I can't see whether my bag has been stolen, I would prefer to be able to have it closer to me so I can touch it, next to me or near my feet. I am sensitive to sounds, so no unnecessary beeping noises if the wheelchair gets too close to things. In the US the tipping culture is huge and I have to prepare a budget for tipping, especially if I will be using these types of services, and for an autonomous wheelchair I wouldn't have to tip.
3	
4	
5	
6	It was hard to get my luggage here, so being able to order the wheelchair to arrive where I need it to would be good (like to my parking). A communication system that tells me where to go or where it's going too.
7	
8	
9	Son: I just want to make sure that my mother is safe and taken care of if she were to travel by herself, that's the most important part for me.
10	
11	If I go to the bathroom with the manual wheelchair, the staff watches my bags, but if I was using the autonomous wheelchair, I would need to take my bags with me into the bathroom and that's tiring. So if it could have something to make sure my bags stay there and are safe from being stolen it would be great.
12	
13	The wheelchair should provide information on flight departures. It should have a map so I don’t get lost in large airports, a speed limit, some sort of horn and a light so that people with hearing impairments can be aware of its presence and avoid accidents. The airport itself should be accessible.

7.2 Appendix B: Staff semi-structured interviews

Question 1

What common difficulties have you encountered working with the wheelchairs?

Interviewee	Answer
1	When the passenger is too heavy the seat bends a bit and it causes the wheelchair to break (slow down). Opening the doors is difficult because I need to hold the door then bring the passenger through it. Also pushing two chairs at the same time, which happens very often, is hard. It's especially hard to make two wheelchairs go through the thin automatic doors at the arrival gate; I have to push one chair first and hold the front part of the second chair to make them both go through.
2	
3	Going through security checks is challenging especially if we need to handle a lot of prms at the same time.
4	Where to put the luggage if the footrest is up. The front wheels sometimes rotate and hit the passenger's front feet.
5	Carrying the luggage at the same time as pushing the passenger if they have a couple luggage is hard, so definitely somewhere to put their luggage.
6	Luggage space is missing.

Question 2

Do you feel like there are any physical elements missing on the wheelchairs?

Interviewee	Answer
1	The footrest often breaks. It might also be due to the fact that that's where we rest the luggage when pushing them.
2	Maybe add some thicker foam padding to the wheelchair seats.
3	From constant use and pushing passengers around, the wheels gradually bend and warp, making it harder to push them. Sometimes the wheelchairs veer off course because of this.
4	Somewhere to put the luggage
5	
6	Something that can take into account the weight of the passenger

Question 3

What are the most common users of the prm service?

Interviewee	Answer
1	
2	
3	
4	
5	senior citizens, those that need assistance and injured people
6	amongst those we also get people with cognitive disabilities or kids that need to be accompanied

Question 4

Any additional comments or needs?

Interviewee	Answer
1	Sometimes the prm passengers are in such a hurry that they decide not to wait for and just leave or the staff has to rush and bring them to their gate.
2	I would say the airport needs some reorganization, in terms of infrastructure too
3	
4	
5	
6	We need new wheelchairs for heavy people.